THE WEIL CONJECTURE. I

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CONTENTS

In this article I prove the Weil conjecture concerning the eigenvalues of Frobenius endomorphisms. A precise statement is given in (1.6). I have tried to present the proof in as geometric and elementary a form as possible, and I have included many reminders: only the results of §§ 3, 6, 7, and 8 are original.

In an article following this one, I shall give various refinements of the intermediate results, and applications, among them the "hard" Lefschetz theorem (on iterated cup products by the cohomology class of a hyperplane section).

The text faithfully follows six lectures given at Cambridge in July 1973. I thank N. Katz for allowing me to use his notes.

1. Grothendieck's theory: cohomological interpretation of L-functions.

(1.1) Let X be a scheme of finite type over **Z**, let |X| be the set of closed points of X, and, for $x \in |X|$, let $\mathrm{N}(x)$ be the number of elements of the residue field $k(x)$ of X at x. The Hasse-Weil zeta function of X is

(1.1.1)
$$\zeta_{\mathrm{X}}(s) = \prod_{x \in |\mathrm{X}|} (1 - \mathrm{N}(x)^{-s})^{-1}$$

(this product converges absolutely for $\mathrm{Re}(s)$ sufficiently large). For $\mathrm{X} = \mathrm{Spec}(\mathbf{Z})$, $\zeta_{\mathrm{X}}(s)$ is the Riemann zeta function.

We shall consider exclusively the case in which X is a scheme over a finite field $\mathbf{F}_q$.

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For $x \in |\mathrm{X}|$, we shall write q_x rather than $\mathrm{N}(x)$. Setting $\deg(x) = [k(x) : \mathbf{F}_q]$, we have $q_x = q^{\deg(x)}$. It is useful to introduce the variable $t = q^{-s}$. Put

(1.1.2)
$$\mathrm{Z}(\mathrm{X}; t) = \prod_{x \in |\mathrm{X}|} (1 - t^{\deg(x)})^{-1};$$

this infinite product converges for $|t|$ sufficiently small, and

(1.1.3)
$$\zeta_{\mathrm{X}}(s) = \mathrm{Z}(\mathrm{X}; q^{-s}).$$

(1.2) Dwork (On the rationality of the zeta function of an algebraic variety, *Amer. J. Math.*, 82, 1960, p. 631-648) and Grothendieck ([1] and SGA 5) proved that $\mathrm{Z}(\mathrm{X}, t)$ is a rational function of t.

For Grothendieck, this is a corollary of general results in ℓ-adic cohomology (ℓ denotes a prime number different from the characteristic p of $\mathbf{F}_q$). These furnish a cohomological interpretation of the zeros and poles of $\mathrm{Z}(\mathrm{X}; t)$, and a functional equation when X is proper and smooth. Dwork's methods are p-adic. For X a nonsingular hypersurface in projective space, they likewise furnished a cohomological interpretation of the zeros and poles, and the functional equation. They inspired the crystalline theory of Grothendieck and Berthelot, which, for X proper and smooth, furnishes a p-adic cohomological interpretation of the zeros and poles, and the functional equation. Building on ideas of Washnitzer, Lubkin created a variant of this theory, valid only for X proper, smooth, and liftable to characteristic zero (A p-adic proof of Weil's conjectures, *Ann. of Math.*, 87, 1968, p. 105-255).

We shall make essential use of Grothendieck's results and recall them below.

(1.3) Let X be an algebraic variety over an algebraically closed field k of characteristic p, i.e. a separated scheme of finite type over k. We do not exclude the case $p = 0$. For every prime $\ell \neq p$, Grothendieck defined ℓ-adic cohomology groups $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$. He also defined cohomology groups with proper support $\mathrm{H}^i_c(\mathrm{X}, \mathbf{Q}_\ell)$. For X proper, the two coincide. The $\mathrm{H}^i_c(\mathrm{X}, \mathbf{Q}_\ell)$ are finite-dimensional vector spaces over $\mathbf{Q}_\ell$, and vanish for $i > 2\dim(\mathrm{X})$.

(1.4) Let X_0 be an algebraic variety over $\mathbf{F}_q$, let $\overline{\mathbf{F}}_q$ be an algebraic closure of $\mathbf{F}_q$, and let X be the algebraic variety over $\overline{\mathbf{F}}_q$ obtained from X_0 by extension of scalars from $\mathbf{F}_q$ to $\overline{\mathbf{F}}_q$. In the language of Weil or Shimura, one would express this situation as: "Let X be an algebraic variety defined over $\mathbf{F}_q$." Let $\mathrm{F} : \mathrm{X} \longrightarrow \mathrm{X}$ be the Frobenius morphism; it sends a point with coordinates x to the point with coordinates x^q; in other words, for $\mathrm{U}_0 a$ Zariski-open subset of X_0 defining an open subset U of X, one has $\mathrm{F}^{-1}(\mathrm{U}) = \mathrm{U}$; for $x \in \mathrm{H}^0(\mathrm{U}_0, \mathcal{O})$, one has $\mathrm{F}^* x = x^q$. Identify the set $|\mathrm{X}|$ of closed points of X with $\mathrm{X}_0(\overline{\mathbf{F}}_q)$ (the set $\mathrm{Hom}_{\mathbf{F}_q}(\mathrm{Spec}(\overline{\mathbf{F}}_q), \mathrm{X}_0)$ of points of X_0 with coordinates in $\overline{\mathbf{F}}_q$), and let

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$\varphi \in \mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ be the Frobenius substitution: $\varphi(x) = x^q$. The action of F on |X| identifies with the action of φ on $\mathrm{X}_0(\overline{\mathbf{F}}_q)$. Consequently:

a) The set X^{F} of closed points of X fixed by F identifies with the set $\mathrm{X}_0(\mathbf{F}_q) \subset \mathrm{X}_0(\overline{\mathbf{F}}_q)$ of points of X defined over $\mathbf{F}_q$. This simply expresses that, for $x \in \overline{\mathbf{F}}_q, x \in \mathbf{F}_q \Leftrightarrow x^q = x$.

b) Likewise, the set $\mathrm{X}^{\mathrm{F}^n}$ of closed points of X fixed by the n-th iterate of F identifies with $\mathrm{X}_0(\mathrm{F}_{q^n})$.

c) The set $|\mathrm{X}_0|$ of closed points of X_0 identifies with the set $|\mathrm{X}|_{\mathrm{F}}$ of F-orbits (or φ-orbits) in |X|. The degree $\deg(x)$ of $x \in |\mathrm{X}_0|$ is the number of elements in the corresponding orbit.

d) From *b*) and *c*) one obtains the formula

(1.4.1)
$$\#\mathrm{X}^{\mathrm{F}^n} = \#\mathrm{X}_0(\mathrm{F}_{q^n}) = \sum_{\deg(x)|n} \deg x$$

(for $x \in |\mathrm{X}_0|$ and $\deg(x)|n, x$ defines $\deg(x)$ points with coordinates in F_{q^n}, all conjugate over $\mathbf{F}_q$).

(1.5) The morphism F is finite, hence in particular proper. It therefore induces maps

$$\mathrm{F}^* : \mathrm{H}^i_c(\mathrm{X}, \mathbf{Q}_\ell) \longrightarrow \mathrm{H}^i_c(\mathrm{X}, \mathbf{Q}_\ell).$$

Grothendieck proved the Lefschetz formula

$$\#\mathrm{X}^{\mathrm{F}} = \sum_i (-1)^i \,\mathrm{Tr}(\mathrm{F}^*, \mathrm{H}^i_c(\mathrm{X}, \mathbf{Q}_\ell));$$

the right-hand side, *a priori* an ℓ-adic number, is an integer and equals the left-hand side. Such a formula is reasonable only because dF=0, even at infinity (X is not assumed proper); the relation dF=0 implies that the fixed points of F have multiplicity one.

An analogous formula holds for the iterates of F:

(1.5.1)
$$\#\mathrm{X}^{\mathrm{F}^n} = \#\mathrm{X}_0(\mathrm{F}_{q^n}) = \sum_i (-1)^i \,\mathrm{Tr}(\mathrm{F}^{*n}, \mathrm{H}^i_c(\mathrm{X}, \mathbf{Q}_\ell)).$$

Take the logarithmic derivative of (1.1.2):

(1.5.2)
$$\begin{aligned} t\frac{d}{dt}\log \mathrm{Z}(\mathrm{X}_0, t) &= \frac{t\frac{d}{dt}\mathrm{Z}(\mathrm{X}_0, t)}{\mathrm{Z}(\mathrm{X}_0, t)} = \sum_{x \in |\mathrm{X}_0|} -\frac{-\deg(x)t^{\deg(x)}}{1 - t^{\deg(x)}} \\ &= \sum_{x\in|\mathrm{X}_0|}\sum_{n>0} \deg(x) t^{n\deg(x)} \underset{(1.4.1)}{=} \sum_n \#\mathrm{X}_0(\mathbf{F}_{q^n})\cdot t^n. \end{aligned}$$

For an endomorphism F of a vector space V, one has the following identity of formal series:

(1.5.3)
$$t\cdot\frac{d}{dt}\log(\det(1 - \mathrm{F}t, \mathrm{V})^{-1}) = \sum_{n>0} \mathrm{Tr}(\mathrm{F}^n, \mathrm{V})t^n$$

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(check this for $\dim(V) = 1$, and observe that both sides are additive in V in a short exact sequence). Substituting (1.5.1) into (1.5.2) and applying (1.5.3), one obtains

$$t \cdot \frac{d}{dt} \log Z(X_0, t)$$
$$= \sum_i (-1)^i t \cdot \frac{d}{dt} \log \det(1 - F^* t, H^i_c(X, \mathbf{Q}_\ell))^{-1},$$

hence

(1.5.4) $$Z(X_0, t) = \prod_i \det(1 - F^* t, H^i_c(X, \mathbf{Q}_\ell))^{(-1)^{i+1}}.$$

The right-hand side is an element of $\mathbf{Q}_\ell(t)$. The formula states that its Taylor expansion at $t = 0$, *a priori* a formal series in $\mathbf{Q}_\ell[[t]]$ with constant term one, belongs to $\mathbf{Z}[[t]]$ and equals the left-hand side, itself also regarded as a formal series in t. This is Grothendieck's cohomological interpretation of the function Z.

Our principal result is the following.

Theorem **(1.6)**. — *Let* X_0 *be a nonsingular (= smooth) projective variety over* $\mathbf{F}_q$. *For every i, the characteristic polynomial* $\det(t \cdot 1 - F^*, H^i(X, \mathbf{Q}_\ell))$ *has integer coefficients independent of* $\ell (\ell \neq p)$. *The complex roots* α *of this polynomial (the complex conjugates of the eigenvalues of* F^**) have absolute value* $|\alpha| = q^{i/2}$.

We first show that (1.6) follows from the apparently weaker result below.

Lemma **(1.7)**. — *For every i and every* $\ell \neq p$, *the eigenvalues of the endomorphism* F^* *of* $H^i(X, \mathbf{Q}_\ell)$ *are algebraic numbers all of whose complex conjugates* α *have absolute value* $|\alpha| = q^{i/2}$.

Proof that (1.7) implies (1.6). — Regard $Z(X_0, t)$ as a formal series with constant term 1, an element of $\mathbf{Z}[[t]]$: $Z(X_0, t) = \sum_n a_n t^n$. By (1.5.3), the image of $Z(X_0, t)$ in $\mathbf{Q}_\ell[[t]]$ is the Taylor expansion of a rational function. This means that, for N and M sufficiently large (at least the degrees of the numerators and denominators), the Hankel determinants

$$H_k = \det((a_{i+j+k})_{0 \le i, j \le M}) (k > N)$$

vanish. This vanishing holds in $\mathbf{Q}_\ell$ if and only if it holds in $\mathbf{Q}$; therefore $Z(X_0, t)$ is the Taylor expansion of an element of $\mathbf{Q}(t)$. In other words,

$$Z(X_0, t) \in \mathbf{Z}[[t]] \cap \mathbf{Q}_\ell(t) \subset \mathbf{Q}(t).$$

Write $Z(X_0, t) = P/Q$, with $P, Q \in \mathbf{Z}[t]$ relatively prime and with positive constant terms. By a lemma of Fatou, the fact that $Z(X_0, t)$ belongs to $\mathbf{Z}[[t]]$ and has constant term one implies that the constant terms of P and Q are 1. Put

$$P_i(t) = \det(1 - F^* t, H^i(X, \mathbf{Q}_\ell)).$$

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By hypothesis (1.7), the P_i are pairwise relatively prime. The right-hand side of (1.5.4) is therefore in reduced form, and

$$\mathrm{P}(t) = \prod_{i\text{odd}} \mathrm{P}_i(t)$$
$$\mathrm{Q}(t) = \prod_{i\text{even}} \mathrm{P}_i(t).$$

Let K be the subfield of an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$ generated over $\mathbf{Q}$ by the roots of $\mathrm{R}(t) = \mathrm{P}(t)\mathrm{Q}(t)$. The roots of $\mathrm{P}_i(t)$ are precisely those roots of $\mathrm{R}(t)$ having the property that all their complex conjugates have absolute value $q^{-i/2}$. This set is stable under $\mathrm{Gal}(\mathrm{K}/\mathbf{Q})$. Hence $\mathrm{P}_i(t)$ has rational coefficients. By Gauss's lemma (or because the roots of P_i, being roots of $\mathrm{R}(t)$, are inverses of algebraic integers), it even has integer coefficients. The preceding description of the roots of $\mathrm{P}_i(t)$ is independent of ℓ; consequently, $\mathrm{P}_i(t)$ itself is independent of ℓ.

The rest of this article is devoted to the proof of (1.7).

(1.8) Grothendieck's theory gives a cohomological interpretation not only of zeta functions, but also of L-functions. The results are as follows.

(1.9) Let X be an algebraic variety over a field k. For the definition of a constructible $\mathbf{Q}_\ell$-sheaf on X, I refer to SGA 5 VI. It will suffice to say that:

a) If $\mathcal{F}$ is a constructible $\mathbf{Q}_\ell$-sheaf on X, there is a finite partition of X into locally closed parts X_i such that $\mathcal{F}|\mathrm{X}_i$ is twisted constant.

b) Suppose X is connected and let $\overline{x}$ be a geometric point of X. For $\mathcal{F}$ twisted constant, $\pi_1(\mathrm{X}, \overline{x})$ acts on the fiber $\mathcal{F}_{\overline{x}}$; the fiber functor at $\overline{x}$ is an equivalence of categories

$$\text{(constructible twisted-constant } \mathbf{Q}_\ell\text{-sheaves on X)} \longmapsto$$
$$\longmapsto \text{(continuous representations of } \pi_1(\mathrm{X}, \overline{x}) \text{ on finite-dimensional } \mathbf{Q}_\ell\text{-vector spaces)}.$$

Such a representation does not, in general, factor through a finite quotient of $\pi_1(\mathrm{X}, \overline{x})$.

c) If $k = \mathbf{C}$, constructible $\mathbf{Q}_\ell$-sheaves on X identify with sheaves of $\mathbf{Q}_\ell$-vector spaces $\mathcal{F}$ on X^{an} such that there is a finite partition of X into Zariski-locally closed parts X_i and, for each i, a local system $\mathcal{F}_i$ of free $\mathbf{Z}_\ell$-modules of finite type on X_i, with

$$\mathcal{F}|\mathrm{X}_i = \mathcal{F}_i \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell.$$

We shall consider only constructible $\mathbf{Q}_\ell$-sheaves and shall simply call them $\mathbf{Q}_\ell$-sheaves.

(1.10) Suppose k is algebraically closed, and let $\mathcal{F}$ be a $\mathbf{Q}_\ell$-sheaf on X. Grothendieck defined ℓ-adic cohomology groups $\mathrm{H}^i(\mathrm{X}, \mathcal{F})$ and $\mathrm{H}^i_c(\mathrm{X}, \mathcal{F})$. The $\mathrm{H}^i_c(\mathrm{X}, \mathcal{F})$ are finite-dimensional vector spaces over $\mathbf{Q}_\ell$ and vanish for $i > 2\dim(\mathrm{X})$.

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For $k = \mathbf{C}$, $\mathrm{H}^i(\mathrm{X},\mathcal{F})$ and $\mathrm{H}^i_c(\mathrm{X},\mathcal{F})$ are the usual cohomology groups (respectively, with proper support) of X^{an}, with coefficients in $\mathcal{F}$.

(1.11) Let X_0 be an algebraic variety over $\mathbf{F}_q$, let X be the corresponding variety over $\overline{\mathbf{F}}_q$, and let $\mathcal{F}_0$ be a sheaf of sets on X_0 (for the étale topology). Denote its inverse image on X by $\mathcal{F}$. Besides the Frobenius morphism $\mathrm{F} : \mathrm{X} \longrightarrow \mathrm{X}$, one then has a canonical isomorphism $\mathrm{F}^* : \mathrm{F}^*\mathcal{F} \xrightarrow{\sim} \mathcal{F}$. Here is a description. Regard $\mathcal{F}_0$ as an étalé space over X_0, i.e. identify $\mathcal{F}_0$ with the algebraic space $[\mathcal{F}_0]$, endowed with an étale morphism $f : [\mathcal{F}_0] \longrightarrow \mathrm{X}_0$, such that $\mathcal{F}_0$ is the sheaf of local sections of $[\mathcal{F}_0]$. The analogous space $[\mathcal{F}]$, étalé over X, is obtained from $[\mathcal{F}_0]$ by extension of scalars. We therefore have a commutative diagram

$$\begin{array}{ccc} [\mathcal{F}] & \xrightarrow{\mathrm{F}} & [\mathcal{F}] \\ \downarrow f & & \downarrow f \\ \mathrm{X} & \xrightarrow{\mathrm{F}} & \mathrm{X} \end{array}$$

whence a morphism $[\mathcal{F}] \longrightarrow \mathrm{X} \times_{\mathrm{F},\mathrm{X},f} [\mathcal{F}] = [\mathrm{F}^*\mathcal{F}]$, which is an isomorphism because f is étale. Its inverse defines the desired isomorphism $\mathrm{F}^*\mathcal{F} \xrightarrow{\sim} \mathcal{F}$.

This construction generalizes to $\mathbf{Q}_\ell$-sheaves.

(1.12) Let X_0 be an algebraic variety over $\mathbf{F}_q$, let $\mathcal{F}_0$ be a $\mathbf{Q}_\ell$-sheaf on X_0, let $(\mathrm{X},\mathcal{F})$ be obtained from $(\mathrm{X}_0,\mathcal{F}_0)$ by extension of scalars from $\mathbf{F}_q$ to $\overline{\mathbf{F}}_q$, and let $\mathrm{F} : \mathrm{X} \longrightarrow \mathrm{X}$ and $\mathrm{F}^* : \mathrm{F}^*\mathcal{F} \longrightarrow \mathcal{F}$.

The finite morphism F and F^* define an endomorphism

$$\mathrm{F}^* : \mathrm{H}^i_c(\mathrm{X},\mathcal{F}) \longrightarrow \mathrm{H}^i_c(\mathrm{X},\mathrm{F}^*\mathcal{F}) \longrightarrow \mathrm{H}^i_c(\mathrm{X},\mathcal{F}).$$

For $x \in |\mathrm{X}|$, F^* defines a morphism $\mathrm{F}^*_x : \mathcal{F}_{\mathrm{F}(x)} \longrightarrow \mathcal{F}_x$. For $x \in \mathrm{X}^{\mathrm{F}}$, this is an endomorphism of $\mathcal{F}_x$. Grothendieck proved the Lefschetz formula

$$\sum_{x\in\mathrm{X}^{\mathrm{F}}} \mathrm{Tr}(\mathrm{F}^*_x,\mathcal{F}_x) = \sum_i (-1)^i \,\mathrm{Tr}(\mathrm{F}^*,\mathrm{H}^i_c(\mathrm{X},\mathcal{F})).$$

An analogous formula holds for the iterates of F: the n-th iterate of F^* defines morphisms $\mathrm{F}^{*n}_x : \mathcal{F}_{\mathrm{F}^n(x)} \longrightarrow \mathcal{F}_x$; for x fixed under F^n, F^{*n}_x is an endomorphism, and

(1.12.1)
$$\sum_{x\in\mathrm{X}^{\mathrm{F}^n}} \mathrm{Tr}(\mathrm{F}^{*n}_x,\mathcal{F}_x) = \sum_i (-1)^i \,\mathrm{Tr}(\mathrm{F}^{*n},\mathrm{H}^i_c(\mathrm{X},\mathcal{F})).$$

(1.13) Let $x_0 \in |\mathrm{X}_0|$, let Z be the corresponding orbit of F in $|\mathrm{X}|$, and let $x \in \mathrm{Z}$. The orbit Z has $\deg(x_0)$ elements (1.4). Denote by $\mathrm{F}^*_{x_0}$ the endomorphism $\mathrm{F}^{*\deg(x_0)}_x$ of $\mathcal{F}_x$, and put

$$\det(1 - \mathrm{F}^*_{x_0}t,\mathcal{F}_0) = \det(1 - \mathrm{F}^*_{x_0}t,\mathcal{F}_x).$$

Up to isomorphism, $(\mathcal{F}_x,\mathrm{F}^*_{x_0})$ does not depend on the choice of x. This justifies omitting x from the notation. We shall use analogous notation for other functions of $(\mathcal{F}_x,\mathrm{F}^*_{x_0})$.

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(1.14) Define $\mathrm{Z}(\mathrm{X}_0, \mathcal{F}_0, t) \in \mathbf{Q}_\ell[[t]]$ by the product

$$\mathrm{Z}(\mathrm{X}_0, \mathcal{F}_0, t) = \prod_{x \in |\mathrm{X}_0|} \det(1 - \mathrm{F}_x^* t^{\deg(x)}, \mathcal{F}_0)^{-1}. \tag{1.14.1}$$

For $\mathcal{F}$ the constant sheaf $\mathbf{Q}_\ell$, this gives (1.1.2). By (1.5.3), the logarithmic derivative of Z is

$$\begin{aligned} t\frac{d}{dt}\log \mathrm{Z}(\mathrm{X}_0, \mathcal{F}_0, t) &\underset{\mathrm{dfn}}{=} \frac{t\frac{d}{dt}\mathrm{Z}(\mathrm{X}_0, \mathcal{F}_0, t)}{\mathrm{Z}(\mathrm{X}_0, \mathcal{F}_0, t)} \\ &= \sum_n \sum_{x \in \mathrm{X}^{\mathrm{F}^n} = \mathrm{X}_0(\mathbf{F}_{q^n})} \mathrm{Tr}(\mathrm{F}_x^{*n}, \mathcal{F}_0) t^n. \end{aligned} \tag{1.14.2}$$

Substituting (1.12.1) into (1.14.2), the same calculation as in (1.5) gives the following generalization of (1.5.4):

$$\mathrm{Z}(\mathrm{X}_0, \mathcal{F}_0, t) = \prod_i \det(1 - \mathrm{F}^* t, \mathrm{H}_c^i(\mathrm{X}, \mathcal{F}))^{(-1)^{i+1}}. \tag{1.14.3}$$

This formula is an identity in $\mathbf{Q}_\ell[[t]]$.

(1.15) It is sometimes convenient to use Galois language rather than geometric language. Here is the dictionary. If $\overline{\mathbf{F}}_q^1$ and $\overline{\mathbf{F}}_q^2$ are two algebraic closures of $\mathbf{F}_q$, then $(\mathrm{X}_0, \mathcal{F}_0)$ over $\mathbf{F}_q$ defines, by extension of scalars, $(\mathrm{X}_1, \mathcal{F}_1)$ over $\overline{\mathbf{F}}_q^1$ and $(\mathrm{X}_2, \mathcal{F}_2)$ over $\overline{\mathbf{F}}_q^2$. Every $\mathbf{F}_q$-isomorphism $\sigma : \overline{\mathbf{F}}_q^1 \xrightarrow{\sim} \overline{\mathbf{F}}_q^2$ induces an isomorphism

$$\mathrm{H}_c^*(\mathrm{X}_1, \mathcal{F}_1) \xrightarrow{\sim} \mathrm{H}_c^*(\mathrm{X}_2, \mathcal{F}_2).$$

In particular, when $\overline{\mathbf{F}}_q^1 = \overline{\mathbf{F}}_q^2$ (denoted $\overline{\mathbf{F}}_q$), one finds that $\mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ acts on $\mathrm{H}_c^*(\mathrm{X}, \mathcal{F})$ (by transport of structure). Let $\varphi \in \mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ be the Frobenius substitution. One checks that

$$\mathrm{F}^* = \varphi^{-1} (\text{ in } \mathrm{End}(\mathrm{H}_c^*(\mathrm{X}, \mathcal{F}))).$$

This leads one to define the *geometric Frobenius* $\mathrm{F} \in \mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ to be φ^{-1}. We have

$$\mathrm{F}^* = \mathrm{F}. \tag{1.15.1}$$

Let x be a geometric point of X_0 localized at $x_0 \in |\mathrm{X}_0|$. By transport of structure, the group $\mathrm{Gal}(k(x)/k(x_0))$ acts on the fiber $(\mathcal{F}_0)_x$ of $\mathcal{F}_0$ at x; in particular, the geometric Frobenius relative to $k(x_0)$, $\mathrm{F}_{x_0} \in \mathrm{Gal}(k(x)/k(x_0))$, acts. When x is defined by a closed point, again denoted x, of X, one has $\mathcal{F}_x = (\mathcal{F}_0)_x$, and

$$\mathrm{F}_{x_0}^* \underset{\mathrm{dfn}}{=} \mathrm{F}_x^{*\deg(x_0)} = \mathrm{F}_{x_0} (\text{ in } \mathrm{End}(\mathcal{F}_x)). \tag{1.15.2}$$

In Galois notation, (1.14.3) is written

$$\begin{aligned} &\prod_{x \in |\mathrm{X}_0|} \det(1 - \mathrm{F}_x t^{\deg(x)}, \mathcal{F}_0)^{-1} \\ &= \prod_i \det(1 - \mathrm{F}t, \mathrm{H}_c^i(\mathrm{X}, \mathcal{F}))^{(-1)^{i+1}}. \end{aligned}$$

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2. Grothendieck's theory: Poincaré duality.

(2.1) To explain the relation between roots of unity and orientations, I shall first restate two classical cases in a fanciful language.

a) *Differentiable manifolds.* — Let X be a differentiable manifold purely of dimension n. The orientation sheaf $\mathbf{Z}'$ on X is the sheaf locally isomorphic to the constant sheaf $\mathbf{Z}$ whose invertible sections over an open subset U of X correspond to the orientations of U. An *orientation* of X is an isomorphism from $\mathbf{Z}'$ to the constant sheaf $\mathbf{Z}$. The *fundamental class* of X is a morphism $\mathrm{Tr} : \mathrm{H}^n_c(\mathrm{X}, \mathbf{Z}') \longrightarrow \mathbf{Z}$; if X is oriented, it identifies with a morphism $\mathrm{Tr} : \mathrm{H}^n_c(\mathrm{X}, \mathbf{Z}) \longrightarrow \mathbf{Z}$. Poincaré duality is expressed with the aid of the fundamental class.

b) *Complex varieties.* — Let $\mathbf{C}$ be an algebraic closure of R. A smooth complex algebraic variety, or rather its underlying differentiable manifold, is always orientable. To orient it, it suffices to orient $\mathbf{C}$ itself. This amounts to the choice:

a) of one of the two roots of the equation $\mathrm{X}^2 = -1$; it is called $+i$;

b) of an isomorphism of $\mathbf{R}/\mathbf{Z}$ with $\mathrm{U}^1 = \{z \in \mathbf{C} | |z| = 1\}$; $+i$ is the image of $1/4$;

c) of one of the two isomorphisms $x \longmapsto \exp(\pm 2\pi i x)$ from $\mathbf{Q}/\mathbf{Z}$ onto the group of roots of unity in $\mathbf{C}$, which extends by continuity to an isomorphism from $\mathbf{R}/\mathbf{Z}$ onto U^1.

Let $\mathbf{Z}(1)$ denote a free $\mathbf{Z}$-module of rank one whose two-element set of generators is in canonical correspondence with one of the two-element sets a), *b*), *c*). The simplest choice is $\mathbf{Z}(1) = \mathrm{Ker}(\exp : \mathbf{C} \longrightarrow \mathbf{C}^*)$. The generator $y = \pm 2\pi i$ corresponds to the isomorphism *c*) $: x \longmapsto \exp(xy)$.

Let $\mathbf{Z}(r)$ be the r-th tensor power of $\mathbf{Z}(1)$. If X is a smooth complex algebraic variety purely of complex dimension r, the orientation sheaf of X is the constant sheaf with value $\mathbf{Z}(r)$.

(2.2) To "orient" algebraic varieties over an algebraically closed field k of characteristic 0, one must choose an isomorphism from $\mathbf{Q}/\mathbf{Z}$ onto the group of roots of unity in k. The set of these isomorphisms is a principal homogeneous space under $\hat{\mathbf{Z}}^*$ (no longer under $\mathbf{Z}^*$). When one is interested only in ℓ-adic cohomology, it suffices to consider roots of unity whose order is a power of ℓ, and to assume that the characteristic p of k is different from ℓ. Let $\mathbf{Z}/\ell^n(1)$ denote the group of roots of unity in k whose order divides ℓ^n. As n varies, the $\mathbf{Z}/\ell^n(1)$ form a projective system, with transition maps

$$\sigma_{m,n} : \mathbf{Z}/\ell^m(1) \longrightarrow \mathbf{Z}/\ell^n(1) : x \longmapsto x^{\ell^{m-n}}.$$

Put $\mathbf{Z}_\ell(1) = \mathrm{lim\,proj}\, \mathbf{Z}/\ell^n(1)$ and $\mathbf{Q}_\ell(1) = \mathbf{Z}_\ell(1) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell$. Let $\mathbf{Q}_\ell(r)$ denote the r-th tensor power of $\mathbf{Q}_\ell(1)$; for $r \in \mathbf{Z}$ negative, put also $\mathbf{Q}_\ell(r) = \mathbf{Q}_\ell(-r)^\vee$.

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As a vector space over $\mathbf{Q}_\ell$, $\mathbf{Q}_\ell(1)$ is isomorphic to $\mathbf{Q}_\ell$. However, the automorphism group of k acts nontrivially on $\mathbf{Q}_\ell(1)$: it acts *through* the character with values in $\mathbf{Z}_\ell^*$ that gives its action on the roots of unity. In particular, if $k = \overline{\mathbf{F}}_q$, the Frobenius substitution $\varphi : x \longmapsto x^q$ acts by multiplication by q.

Let X be a smooth algebraic variety purely of dimension n over k. The *orientation sheaf* of X in ℓ-adic cohomology is the constant $\mathbf{Q}_\ell$-sheaf $\mathbf{Q}_\ell(n)$. The *fundamental class* is a morphism

$$\mathrm{Tr} : \mathrm{H}_c^{2n}(\mathrm{X}, \mathbf{Q}_\ell(n)) \longrightarrow \mathbf{Q}_\ell,$$

or, equivalently,

$$\mathrm{Tr} : \mathrm{H}_c^{2n}(\mathrm{X}, \mathbf{Q}_\ell) \longrightarrow \mathbf{Q}_\ell(-n).$$

Theorem **(2.3)** (Poincaré duality). — *For* X *proper and smooth, purely of dimension n, the bilinear form*

$$\mathrm{Tr}(x \cup y) : \mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell) \otimes \mathrm{H}^{2n-i}(\mathrm{X}, \mathbf{Q}_\ell) \longrightarrow \mathbf{Q}_\ell(-n)$$

is a perfect duality (it identifies $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$ *with the dual of* $\mathrm{H}^{2n-i}(\mathrm{X}, \mathbf{Q}_\ell(n))$*).*

(2.4) Let X_0 be a proper smooth algebraic variety over $\mathbf{F}_q$, purely of dimension n, and let X over $\overline{\mathbf{F}}_q$ be obtained from X_0 by extension of scalars. The morphism (2.3) is compatible with the action of $\mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$. If the (α_j) are the eigenvalues of the geometric Frobenius F acting on $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$, the eigenvalues of F acting on $\mathrm{H}^{2n-i}(\mathrm{X}, \mathbf{Q}_\ell)$ are therefore the $(q^n\alpha_j^{-1})$.

(2.5) Suppose, for simplicity, that X is connected. The proof of (2.4) translates as follows into geometric rather than Galois language (cf. (1.15)).

a) Cup product puts $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$ and $\mathrm{H}^{2n-i}(\mathrm{X}, \mathbf{Q}_\ell)$ in perfect duality with values in $\mathrm{H}^{2n}(\mathrm{X}, \mathbf{Q}_\ell)$, which is one-dimensional.

b) Cup product commutes with pullback F^* by the Frobenius morphism $\mathrm{F} : \mathrm{X} \longrightarrow \mathrm{X}$.

c) The morphism F is finite and of degree q^n: on $\mathrm{H}^{2n}(\mathrm{X}, \mathbf{Q}_\ell)$, F^* is multiplication by q^n.

d) The eigenvalues of F^* therefore have property (2.4).

(2.6) Put $\chi(\mathrm{X}) = \sum_i (-1)^i \dim \mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$. If n is odd, the form $\mathrm{Tr}(x \cup y)$ on $\mathrm{H}^n(\mathrm{X}, \mathbf{Q}_\ell)$ is alternating; the integer $n\chi(\mathrm{X})$ is therefore always even. It follows readily from (1.5.4) and from (2.3), (2.4) that

$$\mathrm{Z}(\mathrm{X}_0, t) = \varepsilon \cdot q^{\frac{-n\chi(\mathrm{X})}{2}} \cdot t^{-\chi(\mathrm{X})} \cdot \mathrm{Z}(\mathrm{X}_0, q^{-n}t^{-1}).$$

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where $\varepsilon = \pm 1$. If n is even, let N denote the multiplicity of the eigenvalue $q^{n/2}$ of F^* acting on $\mathrm{H}^n(\mathrm{X}, \mathbf{Q}_\ell)$ (i.e. the dimension of the corresponding generalized eigenspace). We have

$$\varepsilon = \begin{cases} 1 & \text{if } n \text{ is odd,} \\ (-1)^{\mathrm{N}} & \text{if } n \text{ is even.} \end{cases}$$

This is Grothendieck's formulation of the functional equation for Z-functions.

(2.7) We shall need other forms of the duality theorem. The case of curves would suffice for our use of them. If $\mathcal{F}$ is a $\mathbf{Q}_\ell$-sheaf on an algebraic variety X over an algebraically closed field k, write $\mathcal{F}(r)$ for the sheaf $\mathcal{F} \otimes \mathbf{Q}_\ell(r)$. This sheaf is (noncanonically) isomorphic to $\mathcal{F}$.

Theorem **(2.8)**. — *Suppose* X *is smooth, purely of dimension n, and* $\mathcal{F}$ *is twisted-constant. Let* $\mathcal{F}^\vee$ *be the dual of* $\mathcal{F}$. *The bilinear form*

$$\begin{aligned} \mathrm{Tr}(x \cup y) : \mathrm{H}^i(\mathrm{X}, \mathcal{F}) \otimes \mathrm{H}_c^{2n-i}(\mathrm{X}, \mathcal{F}^\vee(n)) \\ \longrightarrow \mathrm{H}_c^{2n}(\mathrm{X}, \mathcal{F} \otimes \mathcal{F}^\vee(n)) \\ \longrightarrow \mathrm{H}_c^{2n}(\mathrm{X}, \mathbf{Q}_\ell(n)) \longrightarrow \mathbf{Q}_\ell \end{aligned}$$

is a perfect duality.

(2.9) Suppose X is connected and let x be a closed point of X. The functor $\mathcal{F} \longmapsto \mathcal{F}_x$ is an equivalence from the category of twisted-constant $\mathbf{Q}_\ell$-sheaves to the category of ℓ-adic representations of $\pi_1(\mathrm{X}, x)$. *Under* this equivalence, $\mathrm{H}^0(\mathrm{X}, \mathcal{F})$ identifies with the invariants of $\pi_1(\mathrm{X}, x)$ acting on $\mathcal{F}_x$:

(2.9.1) $$\mathrm{H}^0(\mathrm{X}, \mathcal{F}) \xrightarrow{\sim} \mathcal{F}_x^{\pi_1(\mathrm{X},x)}.$$

By (2.8), for X smooth and connected of dimension n, one therefore has

$$\begin{aligned} \mathrm{H}_c^{2n}(\mathrm{X}, \mathcal{F}) &= \mathrm{H}^0(\mathrm{X}, \mathcal{F}^\vee(n))^\vee \\ &= ((\mathcal{F}_x^\vee(n))^{\pi_1(\mathrm{X},x)})^\vee. \end{aligned}$$

Duality exchanges invariants (the largest invariant subspace) and coinvariants (the largest invariant quotient). The formula may therefore be rewritten

$$\mathrm{H}_c^{2n}(\mathrm{X}, \mathcal{F}) = (\mathcal{F}_x)_{\pi_1(\mathrm{X},x)}(-n).$$

We shall use this only for $n = 1$.

Scholium **(2.10)**. — *Let* X *be a smooth connected curve over an algebraically closed field k, let x be a closed point of* X, *and let* $\mathcal{F}$ *be a twisted-constant* $\mathbf{Q}_\ell$*-sheaf. Then*

(i) $\mathrm{H}_c^0(\mathrm{X}, \mathcal{F}) = 0$ *if* X *is affine.*
(ii) $\mathrm{H}_c^2(\mathrm{X}, \mathcal{F}) = (\mathcal{F}_x)_{\pi_1(\mathrm{X},x)}(-1)$.

Assertion (i) simply means that $\mathcal{F}$ has no section with finite support.

(2.11) Let X be a projective, smooth, connected curve over an algebraically closed field k; let U be an open subset of X whose complement is a finite set S of closed points; let j be the inclusion $\mathrm{U} \hookrightarrow \mathrm{X}$; and let $\mathcal{F}$ be a twisted-constant $\mathbf{Q}_\ell$-sheaf on U. Let $j_*\mathcal{F}$ be the constructible $\mathbf{Q}_\ell$-sheaf that is the direct image of $\mathcal{F}$. Its fiber at $x \in \mathrm{S}$ has rank less than or equal to the rank of its fiber at a general point; it is a space of invariants under a local monodromy group.

Theorem **(2.12)**. — *The bilinear form*

$$\begin{aligned}\mathrm{Tr}(x \cup y) : \mathrm{H}^i(\mathrm{X}, j_*\mathcal{F}) \otimes \mathrm{H}^{2-i}(\mathrm{X}, j_*\mathcal{F}^\vee(1)) \\ \longrightarrow \mathrm{H}^2(\mathrm{X}, j_*\mathcal{F} \otimes j_*\mathcal{F}^\vee(1)) \\ \longrightarrow \mathrm{H}^2(\mathrm{X}, j_*(\mathcal{F} \otimes \mathcal{F}^\vee)(1)) \\ \longrightarrow \mathrm{H}^2(\mathrm{X}, j_*\mathbf{Q}_\ell(1)) = \mathrm{H}^2(\mathrm{X}, \mathbf{Q}_\ell(1)) \longrightarrow \mathbf{Q}_\ell\end{aligned}$$

is a perfect duality.

(2.13) It will be convenient for us to have $\mathbf{Q}_\ell$-sheaves $\mathbf{Q}_\ell(r)$ on any scheme X on which ℓ is invertible. The point is to define $\mathbf{Z}/\ell^n(1)$. By definition, $\mathbf{Z}/\ell^n(1)$ is the étale sheaf of ℓ^n-th roots of unity.

(2.14) *Bibliographical notes on §§ 1 and 2.*

A) All important results in étale cohomology are first proved for torsion sheaves. The extension to $\mathbf{Q}_\ell$-sheaves is made by formal passages to the limit. In what follows, for each theorem cited, I shall refer not to a source in which it is proved, but to a source in which its analogue for torsion sheaves is proved.

B) With the exception of the Lefschetz formula and (2.12), all the results in étale cohomology used in this article are proved in SGA 4. For those already stated, the references are: definition of H^i: VII; definition of H^i_c: XVII 5.1; finiteness theorem: XIV 1, completed in XVII 5.3; cohomological dimension: X; Poincaré duality: XVIII.

C) The relation among the various Frobenius elements ((1.4), (1.11), (1.15)) is explained in detail in SGA 5, XV, §§ 1, 2.

D) The cohomological interpretation of Z-functions (1.14.3) is clearly set out in [1]; however, the Lefschetz formula (1.12), for X a projective smooth curve, is used there but not proved. For the proof, one must, alas, refer to SGA 5.

E) The form (2.12) of the Poincaré duality theorem follows from the general result SGA 4, XVIII (3.2.5) (for $\mathrm{S} = \mathrm{Spec}(k), \mathrm{X} = \mathrm{X}, \mathrm{K} = j_*\mathcal{F}, \mathrm{L} = \mathbf{Q}_\ell$) by a local calculation that is not difficult. The statement will appear explicitly in the definitive version of SGA 5. In the case in which we shall use it (tame ramification of $\mathcal{F}$), it could be obtained by transcendental methods, by lifting X and $\mathcal{F}$ to characteristic 0.

3. The fundamental estimate.

The result of this section was catalyzed by reading Rankin [3].

(3.1) Let U_0 be a curve over $\mathbf{F}_q$, the complement in $\mathbf{P}^1$ of a finite set of closed points; let U be the curve over $\overline{\mathbf{F}}_q$ obtained from it; let u be a closed point of U; let $\mathcal{F}_0$ be a twisted-constant $\mathbf{Q}_\ell$-sheaf on U_0; and let $\mathcal{F}$ be its inverse image on U.

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Let $\beta \in \mathbf{Q}$. We shall say that $\mathcal{F}_0$ has *weight* β if, for every $x \in |\mathrm{U}_0|$, the eigenvalues of F_x acting on $\mathcal{F}_0$(1.13) are algebraic numbers all of whose complex conjugates have absolute value $q_x^{\beta/2}$. For example, $\mathbf{Q}_\ell(r)$ has weight $-2r$.

Theorem **(3.2)**. — *Make the following assumptions:*

(i) $\mathcal{F}_0$ *is equipped with a nondegenerate alternating bilinear form*

$$\psi : \mathcal{F}_0 \otimes \mathcal{F}_0 \longrightarrow \mathbf{Q}_\ell(-\beta)(\beta \in \mathbf{Z}).$$

(ii) *The image of* $\pi_1(\mathrm{U}, u)$ *in* $\mathrm{GL}(\mathcal{F}_u)$ *is an open subgroup of the symplectic group* $\mathrm{Sp}(\mathcal{F}_u, \psi_u)$.

(iii) *For every* $x \in |\mathrm{U}_0|$, *the polynomial* $\det(1 - \mathrm{F}_x t, \mathcal{F}_0)$ *has rational coefficients.*

Then $\mathcal{F}$ *has weight* β.

We may suppose, and shall suppose, that U is affine and that $\mathcal{F} \neq 0$.

Lemma **(3.3)**. — *Let* $2k$ *be an even integer, and write* $\overset{2k}{\otimes}\mathcal{F}_0$ *for the* $(2k)$*-th tensor power of* $\mathcal{F}_0$. *For* $x \in |\mathrm{U}_0|$, *the logarithmic derivative*

$$t \cdot \frac{d}{dt} \log(\det(1 - \mathrm{F}_x t^{\deg(x)}, \bigotimes^{2k} \mathcal{F}_0)^{-1})$$

is a formal series with nonnegative rational coefficients.

Hypothesis (iii) ensures that, for every n, $\mathrm{Tr}(\mathrm{F}_x^n, \mathcal{F}_0) \in \mathbf{Q}$. The number

$$\mathrm{Tr}(\mathrm{F}_x^n, \bigotimes^{2k} \mathcal{F}_0) = \mathrm{Tr}(\mathrm{F}_x^n, \mathcal{F}_0)^{2k}$$

is therefore a nonnegative rational number, and we apply (1.5.3).

Lemma **(3.4)**. — *The local factors* $\det(1 - \mathrm{F}_x t^{\deg(x)}, \overset{2k}{\otimes}\mathcal{F}_0)^{-1}$ *are formal series with nonnegative rational coefficients.*

The formal series $\log\det(1 - \mathrm{F}_x t^{\deg(x)}, \overset{2k}{\otimes}\mathcal{F}_0)^{-1}$ has no constant term; by (3.3), its coefficients are ≥ 0; the coefficients of its exponential are therefore also nonnegative.

Lemma **(3.5)**. — *Let* $f_i = \sum_n a_{i,n} t^n$ *be a sequence of formal series with constant term one and nonnegative real coefficients. Suppose that the order of* $f_i - 1$ *tends to infinity with i, and put* $f = \prod_i f_i$. *Then the radius of absolute convergence of* f_i *is at least that of* f.

Indeed, if $f = \sum_n a_n t^n$, then $a_{i,n} \leq a_n$.

Lemma **(3.6)**. — *Under the hypotheses of (3.5), if* f *and the* f_i *are Taylor expansions of meromorphic functions, then*

$$\inf\{|z| \mid f(z) = \infty\} \leq \inf\{|z| \mid f_i(z) = \infty\}.$$

Indeed, these numbers are the radii of absolute convergence.

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(3.7) For each partition P of $[1, 2k]$ into two-element subsets $\{i_\alpha, j_\alpha\}(i_\alpha < j_\alpha)$, define

$$\psi_{\mathrm{P}} : \bigotimes^{2k} \mathcal{F}_0 \longrightarrow \mathbf{Q}_\ell(-k\beta) : x_1 \otimes \cdots \otimes x_{2k} \longmapsto \prod_\alpha \psi(x_{i_\alpha}, x_{j_\alpha}).$$

Let x be a closed point of X. Hypothesis (ii) ensures that the coinvariants of $\pi_1(\mathrm{U}, u)$ in $\overset{2k}{\otimes} \mathcal{F}_u$ are the coinvariants in $\overset{2k}{\otimes} \mathcal{F}_u$ under the full symplectic group (π_1 is Zariski-dense in Sp). Let $\mathcal{P}$ be the set of partitions P. By H. Weyl (*The classical groups*, Princeton University Press, chap. VI, § 1), for a suitable $\mathcal{P}' \subset \mathcal{P}$, depending on $\dim(\mathcal{F}_u)$, the maps ψ_{P} for $\mathrm{P} \in \mathcal{P}'$ define an isomorphism

$$(\bigotimes^{2k} \mathcal{F}_u)_{\pi_1} = (\bigotimes^{2k} \mathcal{F}_u)_{\mathrm{Sp}} \xrightarrow{\sim} \mathbf{Q}_\ell(-k\beta)^{\mathcal{P}'}.$$

Let N be the number of elements of $\mathcal{P}'$. By (2.10), the preceding formula gives

$$\mathrm{H}^2_c(\mathrm{U}, \bigotimes^{2k} \mathcal{F}) \simeq \mathbf{Q}_\ell(-k\beta - 1)^{\mathrm{N}}.$$

Since $\mathrm{H}^0_c(\mathrm{U}, \overset{2k}{\otimes} \mathcal{F}) = 0$, formula (1.14.3) reduces to

$$\mathrm{Z}(\mathrm{U}_0, \bigotimes^{2k} \mathcal{F}_0, t) = \frac{\det(1 - \mathrm{F}^* t, \mathrm{H}^1(\mathrm{U}, \overset{2k}{\otimes} \mathcal{F}))}{(1 - q^{k\beta+1} t)^{\mathrm{N}}}.$$

Thus this function Z is the Taylor expansion of a rational function having no pole except at $t = 1/q^{k\beta+1}$. We shall use only the fact that the poles have absolute value $1/q^{k\beta+1}$ in $\mathbf{C}$. This could be deduced from general arguments about reductive groups. If α is an eigenvalue of F_x on $\mathcal{F}_0$, then α^{2k} is an eigenvalue of F_x on $\overset{2k}{\otimes} \mathcal{F}_0$. Again write α for any complex conjugate of α. The inverse power $1/\alpha^{2k/\deg(x)}$ is a pole of $\det(1 - \mathrm{F}_x t^{\deg(x)}, \overset{2k}{\otimes} \mathcal{F})^{-1}$. By (3.4) and (3.6), we therefore have

$$|1/q^{k\beta+1}| \leq |1/\alpha^{2k/\deg(x)}|,$$

hence

$$|\alpha| \leq q_x^{\frac{\beta}{2} + \frac{1}{2k}}.$$

Letting k tend to infinity, we obtain

$$|\alpha| \leq q_x^{\beta/2}.$$

On the other hand, the existence of ψ ensures that $q_x^\beta \alpha^{-1}$ is also an eigenvalue, whence the inequality

$$|q_x^\beta \alpha^{-1}| \leq q_x^{\beta/2},$$

and therefore

$$q_x^{\beta/2} \leq |\alpha|.$$

This completes the proof.

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Corollary **(3.8)**. — *Let α be an eigenvalue of* F^* *acting on* $\mathrm{H}^1_c(\mathrm{U},\mathcal{F})$. *Then α is an algebraic number all of whose complex conjugates satisfy*

$$|\alpha| \leq q^{\frac{\beta+1}{2}+\frac{1}{2}}.$$

Formula (1.14.3) for $\mathcal{F}_0$ reduces to

$$\mathrm{Z}(\mathrm{U}_0,\mathcal{F}_0,t) = \det(1-\mathrm{F}^*t, \mathrm{H}^1_c(\mathrm{U},\mathcal{F})).$$

The left-hand side is a formal series with rational coefficients, in view of its product expansion and hypothesis (iii). The right-hand side is therefore a polynomial with rational coefficients; $1/\alpha$ is a root. This already proves that α is algebraic. To complete the proof, it is enough to verify that the infinite product defining $\mathrm{Z}(\mathrm{U}_0,\mathcal{F}_0,t)$ converges absolutely (and hence is nonzero) for $|t| < q^{\frac{-\beta}{2}-1}$.

Let N be the rank of $\mathcal{F}$, and write

$$\det(1-\mathrm{F}_x t,\mathcal{F}) = \prod_{i=1}^{\mathrm{N}}(1-\alpha_{i,x}t).$$

By (3.2), $|\alpha_{i,x}| = q_x^{\beta/2}$. The convergence of the infinite product Z follows from that of the series

$$\sum_{i,x} |\alpha_{i,x} t^{\deg(x)}|.$$

For $|t| = q^{\frac{-\beta}{2}-1-\varepsilon}$ $(\varepsilon > 0)$, we have

$$\sum_{i,x} |\alpha_{i,x} t^{\deg(x)}| = \mathrm{N}\sum_x q_x^{-1-\varepsilon}.$$

On the affine line there are q^n points with values in $\mathbf{F}_{q^n}$, and hence at most q^n closed points of degree n. Thus

$$\sum_x q_x^{-1-\varepsilon} \leq \sum_n q^n \cdot q^{n(-1-\varepsilon)} = \sum_n q^{-n\varepsilon} < \infty,$$

which completes the proof.

Corollary **(3.9)**. — *Let j_0 be the inclusion of* U_0 *in* $\mathbf{P}^1_{\mathbf{F}_q}$, *let j be the inclusion of* U *in* $\mathbf{P}^1$, *and let α be an eigenvalue of* F^* *acting on* $\mathrm{H}^1(\mathbf{P}^1, j_*\mathcal{F})$. *Then α is an algebraic number all of whose complex conjugates satisfy*

$$q^{\frac{\beta+1}{2}-\frac{1}{2}} \leq |\alpha| \leq q^{\frac{\beta+1}{2}+\frac{1}{2}}.$$

A segment of the long exact cohomology sequence defined by the short exact sequence

$$0 \longrightarrow j_!\mathcal{F} \longrightarrow j_*\mathcal{F} \longrightarrow j_*\mathcal{F}/j_!\mathcal{F} \longrightarrow 0$$

($j_!$ means extension by zero) is

$$\mathrm{H}^1_c(\mathrm{U},\mathcal{F}) \longrightarrow \mathrm{H}^1(\mathbf{P}^1, j_*\mathcal{F}) \longrightarrow 0.$$

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The eigenvalue α therefore already occurs in $\mathrm{H}_c^1(\mathrm{U},\mathcal{F})$, and (3.8) applies to it:

$$|\alpha| \leq q^{\frac{\beta+1}{2}+\frac{1}{2}}.$$

Poincaré duality (2.12) ensures that $q^{\beta+1}\alpha^{-1}$ is also an eigenvalue, whence

$$|q^{\beta+1}\alpha^{-1}| \leq q^{\frac{\beta+1}{2}+\frac{1}{2}},$$

and the corollary follows.

4. Lefschetz theory: local theory.

(4.1) Over **C**, the local results of Lefschetz are as follows.

Let $\mathrm{D} = \{z \mid |z| < 1\}$ be the unit disk, let $\mathrm{D}^* = \mathrm{D} - \{0\}$, and let $f : \mathrm{X} \longrightarrow \mathrm{D}$ be a morphism of analytic spaces. Suppose that

a) X is nonsingular and purely of dimension $n+1$;

$b)$ f is proper;

$c)$ f is smooth away from a point x of the special fiber $\mathrm{X}_0 = f^{-1}(0)$;

$d)$ at x, f has a nondegenerate quadratic point.

Let $t \neq 0$ in D and let $\mathrm{X}_t = f^{-1}(t)$ be "the" general fiber. To the preceding data one associates:

α) specialization morphisms $sp : \mathrm{H}^i(\mathrm{X}_0, \mathbf{Z}) \longrightarrow \mathrm{H}^i(\mathrm{X}_t, \mathbf{Z})$: X_0 is a deformation retract of X, and sp is the composite arrow

$$\mathrm{H}^i(\mathrm{X}_0, \mathbf{Z}) \xleftarrow{\sim} \mathrm{H}^i(\mathrm{X}, \mathbf{Z}) \longrightarrow \mathrm{H}^i(\mathrm{X}_t, \mathbf{Z});$$

β) monodromy transformations $\mathrm{T} : \mathrm{H}^i(\mathrm{X}_t, \mathbf{Z}) \longrightarrow \mathrm{H}^i(\mathrm{X}_t, \mathbf{Z})$, which describe the effect on the singular cycles of X_t of "turning t around 0." Equivalently, this is the action on $\mathrm{H}^i(\mathrm{X}_t, \mathbf{Z})$, the fiber at t of the local system $\mathrm{R}^i f_* \mathbf{Z} | \mathrm{D}^*$, of the positive generator of $\pi_1(\mathrm{D}^*, t)$.

Lefschetz theory describes α) and β) in terms of the *vanishing cycle* $\delta \in \mathrm{H}^n(\mathrm{X}_t, \mathbf{Z})$. This cycle is well defined up to sign. For $i \neq n, n+1$, one has

$$\mathrm{H}^i(\mathrm{X}_0, \mathbf{Z}) \xrightarrow{\sim} \mathrm{H}^i(\mathrm{X}_t, \mathbf{Z}) (i \neq n, n+1).$$

For $i = n, n+1$, one has an exact sequence

$$\begin{aligned} 0 \longrightarrow \mathrm{H}^n(\mathrm{X}_0, \mathbf{Z}) \longrightarrow \mathrm{H}^n(\mathrm{X}_t, \mathbf{Z}) &\xrightarrow{x \mapsto (x,\delta)} \mathbf{Z} \\ \longrightarrow \mathrm{H}^{n+1}(\mathrm{X}_0, \mathbf{Z}) \longrightarrow \mathrm{H}^{n+1}(\mathrm{X}_t, \mathbf{Z}) &\longrightarrow 0. \end{aligned}$$

For $i \neq n$, the monodromy T is the identity. For $i = n$,

$$\mathrm{T}x = x \pm (x, \delta)\delta.$$

The values of this sign, of $\mathrm{T}\delta$, and of (δ, δ) are as follows:

n mod 4	0	1	2	3
$\mathrm{T}x = x \pm (x,\delta)\delta$	$-$	$-$	$+$	$+$
(δ,δ)	2	0	-2	0
$\mathrm{T}\delta$	$-\delta$	δ	$-\delta$	δ

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The monodromy transformation T preserves the intersection form $\mathrm{Tr}(x \cup y)$ on $\mathrm{H}^n(\mathrm{X}_t, \mathbf{Z})$. For n odd, it is a symplectic transvection. For n even, it is an orthogonal reflection.

(4.2) Here is the analogue of (4.1) in abstract algebraic geometry. The disk D is replaced by the spectrum of a henselian discrete valuation ring A with algebraically closed residue field. Let S be this spectrum, η its generic point (the spectrum of the fraction field of A), and s its closed point (the spectrum of the residue field). The role of t is played by a geometric generic point $\overline{\eta}$ (the spectrum of an algebraic closure of the fraction field of A).

Let $f : \mathrm{X} \longrightarrow \mathrm{S}$ be a proper morphism, with X regular and purely of dimension $n+1$. Suppose f is smooth except for an ordinary quadratic point x in the special fiber X_s. Let ℓ be a prime number different from the residual characteristic p of S. Writing $\mathrm{X}_{\overline{\eta}}$ for the geometric generic fiber, one again has a specialization morphism

(4.2.1)
$$sp : \mathrm{H}^i(\mathrm{X}_s, \mathbf{Q}_\ell) \xleftarrow{\sim} \mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell) \longrightarrow \mathrm{H}^i(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell).$$

The role of T is played by the action of the inertia group $\mathrm{I} = \mathrm{Gal}(\overline{\eta}/\eta)$ on $\mathrm{H}^i(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell)$ by transport of structure (cf. (1.15)):

(4.2.2)
$$\mathrm{I} = \mathrm{Gal}(\overline{\eta}/\eta) \longrightarrow \mathrm{GL}(\mathrm{H}^i(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell)).$$

The data (4.2.1), (4.2.2) completely describe the sheaves $\mathrm{R}^i f_* \mathbf{Q}_\ell$ on S.

(4.3) Put $n = 2m$ for n even and $n = 2m+1$ for n odd. The data (4.2.1) and (4.2.2) may again be described in terms of a vanishing cycle

(4.3.1)
$$\delta \in \mathrm{H}^n(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell)(m).$$

This cycle is well defined up to sign.

For $i \neq n, n+1$, one has

(4.3.2)
$$\mathrm{H}^i(\mathrm{X}_s, \mathbf{Q}_\ell) \xrightarrow{\sim} \mathrm{H}^i(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell) (i \neq n, n+1).$$

For $i = n, n+1$, one has an exact sequence

(4.3.3)
$$\begin{gathered} 0 \longrightarrow \mathrm{H}^n(\mathrm{X}_s, \mathbf{Q}_\ell) \longrightarrow \mathrm{H}^n(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell) \\ \xrightarrow{x \mapsto \mathrm{Tr}(x \cup \delta)} \mathbf{Q}_\ell(m-n) \longrightarrow \mathrm{H}^{n+1}(\mathrm{X}_s, \mathbf{Q}_\ell) \\ \longrightarrow \mathrm{H}^{n+1}(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell) \longrightarrow 0. \end{gathered}$$

The action (4.2.2) of I (local monodromy) is trivial if $i \neq n$. For $i = n$, it is described as follows.

A) *n odd*. — There is a canonical homomorphism

$$t_\ell : \mathrm{I} \longrightarrow \mathbf{Z}_\ell(1),$$

and the action of $\sigma \in \mathrm{I}$ is

$$x \longmapsto x \pm t_\ell(\sigma)(x, \delta)\delta.$$

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B) *n even*. — We shall not need this case. Let us merely say that, if $p \neq 2$, there is a unique character of order two

$$\varepsilon : \mathrm{I} \longrightarrow \{\pm 1\},$$

and

$$\begin{array}{ll} \sigma x = x & \text{if } \varepsilon(\sigma) = 1, \\ \sigma x = x \pm (x, \delta)\delta & \text{if } \varepsilon(\sigma) = -1. \end{array}$$

The signs $\pm$ in A) and B) are the same as in (4.1).

(4.4) These results give the following information about the sheaves $\mathrm{R}^i f_* \mathbf{Q}_\ell$.

a) *If* $\delta \neq 0$:

1) For $i \neq n$, the sheaf $\mathrm{R}^i f_* \mathbf{Q}_\ell$ is constant.

2) Let j be the inclusion of η in S. Then

$$\mathrm{R}^n f_* \mathbf{Q}_\ell = j_* j^* \mathrm{R}^n f_* \mathbf{Q}_\ell.$$

b) *If* $\delta = 0$: (This is an exceptional case. Since $(\delta, \delta) = \pm 2$ for n even, it can occur only for n odd.)

1) For $i \neq n + 1$, the sheaf $\mathrm{R}^i f_* \mathbf{Q}_\ell$ is constant.

2) Let $\mathbf{Q}_\ell(m - n)_s$ be the sheaf $\mathbf{Q}_\ell(m - n)$ on $\{s\}$, extended by zero on S. There is an exact sequence

$$\begin{gathered} 0 \longrightarrow \mathbf{Q}_\ell(m - n)_s \longrightarrow \mathrm{R}^{n+1} f_* \mathbf{Q}_\ell \\ \longrightarrow j_* j^* \mathrm{R}^{n+1} f_* \mathbf{Q}_\ell \longrightarrow 0, \end{gathered}$$

where $j_* j^* \mathrm{R}^{n+1} f_* \mathbf{Q}_\ell$ is a constant sheaf.

5. Lefschetz theory: global theory.

(5.1) Over **C**, the results of Lefschetz are as follows. Let **P** be a projective space of dimension ≥ 1, and let $\check{\mathbf{P}}$ be the dual projective space; its points parametrize the hyperplanes of **P**, and write H_t for the hyperplane defined by $t \in \check{\mathbf{P}}$. If A is a linear subspace of **P** of codimension 2, the hyperplanes containing A are parametrized by the points of a line $\mathrm{D} \subset \check{\mathbf{P}}$, the *dual* of A. These hyperplanes $(\mathrm{H}_t)_{t \in \mathrm{D}}$ form the *pencil with axis* A.

Let $\mathrm{X} \subset \mathbf{P}$ be a connected nonsingular projective variety of dimension $n + 1$. Let $\widetilde{\mathrm{X}} \subset \mathrm{X} \times \mathrm{D}$ be the set of pairs (x, t) such that $x \in \mathrm{H}_t$. The first and second projections form a diagram

(5.1.1)
$$\begin{array}{ccc} \mathrm{X} & \xleftarrow{\pi} & \widetilde{\mathrm{X}} \\ & & \downarrow f \\ & & \mathrm{D} \end{array}$$

The fiber of f at $t \in \mathrm{D}$ is the hyperplane section $\mathrm{X}_t = \mathrm{X} \cap \mathrm{H}_t$ of X.

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Fix X, and take A sufficiently general. Then:

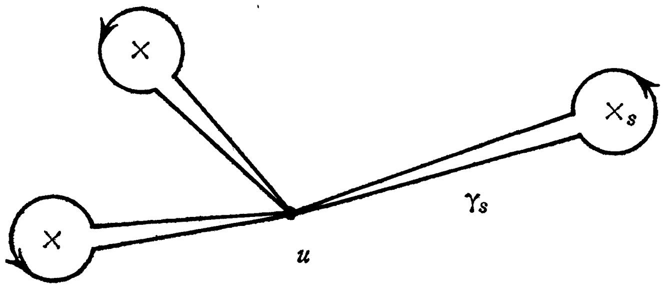
A)A is transverse to X, and $\widetilde{\mathrm{X}}$ is obtained from X by blowing up A ∩ X. In particular, $\widetilde{\mathrm{X}}$ is nonsingular.

B) There is a finite subset S of D and, for each $s \in \mathrm{S}$, a point $x_s \in \mathrm{X}_s$, such that f is smooth away from the x_s.

C) The x_s are nondegenerate critical points of f.

For each $s \in \mathrm{S}$, the local Lefschetz theory of (4.1) therefore applies to a small disk D_s about s and to $f^{-1}(\mathrm{D}_s)$.

(5.2) Put $\mathrm{U} = \mathrm{D} - \mathrm{S}$. Let $u \in \mathrm{U}$, and choose disjoint loops $(\gamma_s)_{s\in\mathrm{S}}$ starting at u, with γ_s going once around s:



These loops generate the fundamental group $\pi_1(\mathrm{U}, u)$. This group acts on $\mathrm{H}^i(\mathrm{X}_u, \mathbf{Z})$, the fiber at u of the local system $\mathrm{R}^i f_* \mathbf{Z}|\mathrm{U}$. By the local theory (4.1), to each $s \in \mathrm{S}$ there corresponds a vanishing cycle $\delta_s \in \mathrm{H}^n(\mathrm{X}_u, \mathbf{Z})$; these cycles depend on the choice of the γ_s. For $i \neq n$, the action of $\pi_1(\mathrm{U}, u)$ on $\mathrm{H}^i(\mathrm{X}_u, \mathbf{Z})$ is trivial. For $i = n$, one has

(5.2.1) $$\gamma_s x = x \pm (x, \delta_s)\delta_s.$$

Let E be the subspace of $\mathrm{H}^n(\mathrm{X}_u, \mathbf{Q})$ spanned by the δ_s (the *vanishing part* of the cohomology).

Proposition **(5.3)**. — E *is stable under the action of the monodromy group* $\pi_1(\mathrm{U}, u)$. *The orthogonal complement* $\mathrm{E}^\perp$ *of* E *(for the intersection form* $\mathrm{Tr}(x \cup y)$*) is the subspace of monodromy invariants in* $\mathrm{H}^n(\mathrm{X}_u, \mathbf{Q})$.

Since the γ_s generate the monodromy group, this is clear from (5.2.1).

Theorem **(5.4)**. — *The vanishing cycles* $\pm\delta_s$ *(taken up to sign) are conjugate under the action of* $\pi_1(\mathrm{U}, u)$.

Let $\check{\mathrm{X}} \subset \check{\mathbf{P}}$ be the dual variety of X: it is the set of $t \in \check{\mathbf{P}}$ such that H_t is tangent to X, i.e. such that X_t is singular, or $\mathrm{X} \subset \mathrm{H}_t$. The variety $\check{\mathrm{X}}$ is irre-

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ducible. Let $\mathrm{Y} \subset \mathrm{X} \times \check{\mathbf{P}}$ be the space of pairs (x, t) such that $x \in \mathrm{H}_t$. There is a diagram

$$\begin{array}{ccc} \mathrm{X} & \leftarrow & \mathrm{Y} \\ & & \downarrow g \\ & & \check{\mathbf{P}} \end{array}$$

The fiber of g at $t \in \check{\mathbf{P}}$ is the hyperplane section $\mathrm{X}_t = \mathrm{X} \cap \mathrm{H}_t$ of X, and g is smooth away from the inverse image of $\check{\mathrm{X}}$.

We recover the situation of (5.1) by replacing $\check{\mathbf{P}}$ by a line $\mathrm{D} \subset \check{\mathbf{P}}$ and Y by $g^{-1}(\mathrm{D})$. Then $\mathrm{S} = \mathrm{D} \cap \check{\mathrm{X}}$. By a theorem of Lefschetz, for D sufficiently general the map

$$\pi_1(\mathrm{D} - \mathrm{S}, u) \longrightarrow \pi_1(\check{\mathbf{P}} - \check{\mathrm{X}}, u)$$

is surjective. It is therefore enough to show that the $\pm\delta_s$ are conjugate under $\pi_1(\check{\mathbf{P}} - \check{\mathrm{X}})$.

For x in the smooth codimension-one locus of $\check{\mathrm{X}}$, let ch be a path from t to x in $\check{\mathbf{P}} - \check{\mathrm{X}}$, and let γ_x be the loop which follows ch to a neighborhood of $\check{\mathrm{X}}$, goes once around $\check{\mathrm{X}}$, and then returns to t along ch. The loops γ_x (as ch varies) are conjugate to one another. Since $\check{\mathrm{X}}$ is irreducible, any two points of its smooth locus can be joined within $\check{\mathrm{X}}$ by a path which remains in the smooth locus. It follows that the conjugacy class of γ_x is independent of x. In particular, the γ_s are conjugate to one another. Formula (5.2.1) then shows that the $\pm\delta_s$ are conjugate.

Corollary **(5.5)**. — *The action of* $\pi_1(\mathrm{U}, u)$ *on* $\mathrm{E}/(\mathrm{E} \cap \mathrm{E}^{\perp})$ *is absolutely irreducible.*

Let $\mathrm{F} \subset \mathrm{E} \otimes \mathbf{C}$ be a subspace stable under monodromy. If $\mathrm{F} \not\subset (\mathrm{E} \cap \mathrm{E}^{\perp}) \otimes \mathbf{C}$, there exist $x \in \mathrm{F}$ and $s \in \mathrm{S}$ such that $(x, \delta_s) \neq 0$. Then

$$\gamma_s x - x = \pm(x, \delta_s)\delta_s \in \mathrm{F},$$

and $\delta_s \in \mathrm{F}$. By (5.4), all the δ_s then belong to F, and $\mathrm{F} = \mathrm{E}$. This proves (5.5).

(5.6) These results transpose as follows to abstract algebraic geometry.

Let $\mathbf{P}$ be a projective space of dimension >1 over an algebraically closed field k of characteristic p, and let $\mathrm{X} \subset \mathbf{P}$ be a connected nonsingular projective variety of dimension $n + 1$. For a linear subspace A of $\mathbf{P}$ of codimension 2, define D, the pencil $(\mathrm{H}_t)_{t \in \mathrm{D}}$, the X_t, $\widetilde{\mathrm{X}}$, and diagram (5.1.1) as in (5.1). The $(\mathrm{X}_t)_{t \in \mathrm{D}}$ are said to form a *Lefschetz pencil* of hyperplane sections if the following conditions hold:

A) The axis A is transverse to X. The space $\widetilde{\mathrm{X}}$ is then obtained from X by blowing up $\mathrm{A} \cap \mathrm{X}$, and is smooth.

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B) There is a finite subset S of D and, for each $s \in \mathrm{S}$, a point $x_s \in \mathrm{X}_s$, such that f is smooth away from the x_s.

C)x_s is an ordinary quadratic singular point of X_s.

For each $s \in \mathrm{S}$, the local Lefschetz theory of § 4 applies to the spectrum D_s of the henselization of the local ring of D at s, and to $\widetilde{\mathrm{X}}_{\mathrm{D}_s} = \widetilde{\mathrm{X}} \times_{\mathrm{D}} \mathrm{D}_s$.

(5.7) Let N be the dimension of **P**, let r be an integer ≥ 1, and let $\iota_{(r)}$ be the embedding of **P** into projective space of dimension $\binom{\mathrm{N}+r}{\mathrm{N}} - 1$ whose homogeneous coordinates are the monomials of degree r in the homogeneous coordinates of **P**. The hyperplane sections of $\iota_{(r)}(\mathbf{P})$ are the hypersurfaces of degree r in **P**.

If $p \neq 0$, it may happen that no pencil of hyperplane sections of X is a Lefschetz pencil. Nevertheless, if $r \geq 2$ and the given projective embedding $\iota_1 : \mathrm{X} \hookrightarrow \mathbf{P}$ is replaced by $\iota_r = \iota_{(r)} \circ \iota_1$, then, in this new embedding, every sufficiently general pencil is a Lefschetz pencil. In other words, if $r \geq 2$, a sufficiently general pencil of hypersurface sections of degree r of X is always a Lefschetz pencil.

(5.8) In the remainder of this discussion we study a Lefschetz pencil of hyperplane sections of X, *excluding the case* $p = 2$ *with* n *even*. The case in which n is odd would suffice for what follows. Put $\mathrm{U} = \mathrm{D} - \mathrm{S}$. Let $u \in \mathrm{U}$ and let ℓ be a prime $\neq p$. The local results of § 4 show that $\mathrm{R}^n f_* \mathbf{Q}_\ell$ is *tamely* ramified at each $s \in \mathrm{S}$. The tame fundamental group of U is a quotient of the profinite completion of the analogous transcendental fundamental group (lifting tame coverings to characteristic 0, and the Riemann existence theorem). The algebraic situation is therefore entirely analogous to the transcendental one, and the Lefschetz results transpose by standard arguments. In the proof of (5.4), the Lefschetz theorem on fundamental groups becomes Bertini's theorem, and one must invoke Abhyankar's lemma to control the ramification of $\mathrm{R}^i g_* \mathbf{Q}_\ell$ along the smooth codimension-one locus of $\check{\mathrm{X}}$.

The results are as follows.

a) *If the vanishing cycles are nonzero:*

1) For $i \neq n$, the sheaf $\mathrm{R}^i f_* \mathbf{Q}_\ell$ on D is constant.

2) Let j be the inclusion of U in D. Then

$$\mathrm{R}^n f_* \mathbf{Q}_\ell = j_* j^* \mathrm{R}^n f_* \mathbf{Q}_\ell.$$

3) Let $\mathrm{E} \subset \mathrm{H}^n(\mathrm{X}_u, \mathbf{Q}_\ell)$ be the subspace of cohomology spanned by the vanishing cycles. This subspace is stable under $\pi_1(\mathrm{U}, u)$, and

$$\mathrm{E}^\perp = \mathrm{H}^n(\mathrm{X}_u, \mathbf{Q}_\ell)^{\pi_1(\mathrm{U},u)}.$$

The representation of $\pi_1(\mathrm{U}, u)$ on $\mathrm{E}/(\mathrm{E} \cap \mathrm{E}^\perp)$ is absolutely irreducible, and the image of π_1 in $\mathrm{GL}(\mathrm{E}/(\mathrm{E} \cap \mathrm{E}^\perp))$ is generated (topologically) by the maps $x \longmapsto x \pm (x, \delta_s)\delta_s (s \in \mathrm{S})$, with the sign $\pm$ determined as in (4.1).

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b) *If the vanishing cycles are zero:* (This is an exceptional case. Since $(\delta,\delta) = \pm 2$ for n even, it can occur only when n is odd: $n = 2m+1$. Note that if one vanishing cycle is zero, then all are zero, since they are conjugate.)

1) For $i \neq n+1$, the sheaf $\mathrm{R}^i f_* \mathbf{Q}_\ell$ is constant.

2) There is an exact sequence

$$0 \longrightarrow \bigoplus_{s\in \mathrm{S}} \mathbf{Q}_\ell(m-n)_s \longrightarrow \mathrm{R}^{n+1} f_* \mathbf{Q}_\ell \longrightarrow \mathcal{F} \longrightarrow 0$$

with $\mathcal{F}$ constant.

3)$\mathrm{E} = 0$.

(5.9) The subspace $\mathrm{E} \cap \mathrm{E}^\perp$ of E is the kernel of the restriction to E of the intersection form $\mathrm{Tr}(x \cup y)$. This form therefore induces a nondegenerate bilinear form

$$\psi : \mathrm{E}/(\mathrm{E}\cap \mathrm{E}^\perp) \otimes \mathrm{E}/(\mathrm{E}\cap \mathrm{E}^\perp) \longrightarrow \mathbf{Q}_\ell(-n),$$

alternating for n odd and symmetric for n even. This form is preserved by monodromy; for n odd, the monodromy representation therefore induces

$$\rho : \pi_1(\mathrm{U}, u) \longrightarrow \mathrm{Sp}(\mathrm{E}/(\mathrm{E}\cap \mathrm{E}^\perp), \psi).$$

Theorem **(5.10)** (Kajdan-Margulis). — *The image of ρ is open.*

The image of ρ is a compact, hence ℓ-adic analytic, subgroup of $\mathrm{Sp}(\mathrm{E}/(\mathrm{E}\cap \mathrm{E}^\perp), \psi)$. It is enough to show that its Lie algebra $\mathfrak{L}$ is equal to $\mathfrak{sp}(\mathrm{E}/(\mathrm{E}\cap \mathrm{E}^\perp), \psi)$. The transcendental analogue of this Lie algebra is the Lie algebra of the Zariski closure of the monodromy group.

It follows from (5.8) that $\mathfrak{L}$ is generated by the square-zero transformations

$$\mathrm{N}_s : x \longmapsto (x, \delta_s)\delta_s (s \in \mathrm{S})$$

and that $\mathrm{E}/(\mathrm{E}\cap \mathrm{E}^\perp)$ is an absolutely irreducible representation of $\mathfrak{L}$. The theorem follows from the next lemma.

Lemma **(5.11)**. — *Let* V *be a finite-dimensional vector space over a field k of characteristic 0, let ψ be a nondegenerate alternating form, and let $\mathfrak{L}$ be a Lie subalgebra of* $\mathfrak{sp}(\mathrm{V}, \psi)$*. Suppose that:*

(i) V *is a simple representation of* $\mathfrak{L}$*.*

(ii) $\mathfrak{L}$ *is generated by a family of endomorphisms of* V *of the form* $x \longmapsto \psi(x,\delta)\delta$*.*

Then $\mathfrak{L} = \mathfrak{sp}(\mathrm{V}, \psi)$*.*

We may and shall suppose that V, and hence $\mathfrak{L}$, is nonzero. Let $\mathrm{W} \subset \mathrm{V}$ be the set of $\delta \in \mathrm{V}$ such that $\mathrm{N}(\delta) : x \longmapsto \psi(x,\delta)\delta$ belongs to $\mathfrak{L}$.

a) W is stable under homotheties (since $\mathfrak{L}$ is a vector subspace of $\mathfrak{gl}(\mathrm{V})$).

b) If $\delta \in \mathrm{W}$, $\exp(\lambda \mathrm{N}(\delta))$ is an automorphism of $(\mathrm{V}, \psi, \mathfrak{L})$, and therefore carries W into itself. If $\delta', \delta'' \in \mathrm{W}$, then $\exp(\lambda \mathrm{N}(\delta'))\cdot \delta'' = \delta'' + \lambda\psi(\delta'',\delta')\delta' \in \mathrm{W}$; if $\psi(\delta',\delta'') \neq 0$, the linear subspace spanned by δ' and δ'' lies in W.

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c) It follows that W is the union of its maximal linear subspaces W_α, and that these are pairwise orthogonal. Each W_α is therefore stable under the $\mathrm{N}(\delta)(\delta \in \mathrm{W})$, hence stable under $\mathfrak{L}$. By hypothesis (i), $\mathrm{W}_\alpha = \mathrm{V}$, and $\mathfrak{L}$ contains every $\mathrm{N}(\delta)$ for $\delta \in \mathrm{V}$. We conclude by observing that the Lie algebra $\mathfrak{sp}(\mathrm{V}, \psi)$ is generated by the $\mathrm{N}(\delta)(\delta \in \mathrm{V})$.

Remark **(5.12)** (not used below). — It is now easy to prove (1.6) for a hypersurface of odd dimension n in $\mathbf{P}^{n+1}_{\mathbf{F}_q}$.

Let X_0 be such a hypersurface, and let $\overline{\mathrm{X}}_0$ be the hypersurface over $\overline{\mathbf{F}}_q$ obtained from it by extension of scalars. One has

$$\mathrm{H}^{2i}(\overline{\mathrm{X}}_0, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-i)(0 \leq i \leq n);$$

$\mathrm{H}^{2i}(\overline{\mathrm{X}}_0, \mathbf{Q}_\ell(i))$ is generated by the i-th cup power of η, the cohomology class $c_1(\mathcal{O}(1))$ of a hyperplane section. Hence

$$\mathrm{Z}(\mathrm{X}_0, t) = \det(1 - \mathrm{F}^* t, \mathrm{H}^n(\overline{\mathrm{X}}_0, \mathbf{Q}_\ell)) / \prod_{i=0}^{n} (1 - q^i t),$$

and $\det(1 - \mathrm{F}^* t, \mathrm{H}^n(\overline{\mathrm{X}}_0, \mathbf{Q}_\ell))$ is a polynomial with integer coefficients independent of ℓ.

Let X_0 vary in a Lefschetz pencil of hypersurfaces defined over $\mathbf{F}_q$ (cf. (5.7) for $\mathrm{X} = \mathbf{P}^{n+1}$; the existence of such a pencil is not clear; to complete the argument sketched here one would have to use the arguments to be given in (7.1)). One checks that here E coincides with the whole H^n, and (3.2) gives the Weil conjecture for every hypersurface in the pencil, in particular for X_0.

(5.13) *Bibliographical indications for §§ 4 and 5.*

A) Lefschetz's results (4.1) and (5.1) through (5.5) are contained in his book [2]. For the local theory (4.1), it may be more convenient to consult SGA 7, XIV (3.2).

B) The results of § 4 are proved in Exposés XIII, XIV, and XV of SGA 7.

C) (5.7) is proved in SGA 7, XVII.

D) (5.8) is proved in SGA 7, XVIII. The irreducibility theorem is proved there for E, but only under the hypothesis $\mathrm{E} \cap \mathrm{E}^\perp = \{0\}$. The proof in the general case, for $\mathrm{E}/(\mathrm{E} \cap \mathrm{E}^\perp)$, is the same.

6. A rationality theorem.

(6.1) Let $\mathbf{P}_0$ be a projective space of dimension ≥ 1 over $\mathbf{F}_q$, let $\mathrm{X}_0 \subset \mathbf{P}_0$ be a nonsingular projective variety, let $\mathrm{A}_0 \subset \mathbf{P}_0$ be a linear subspace of codimension two, let $\mathrm{D}_0 \subset \check{\mathbf{P}}_0$ be the dual line, let $\overline{\mathbf{F}}_q$ be an algebraic closure of $\mathbf{F}_q$, and let $\mathbf{P}$, X, A, D over $\overline{\mathbf{F}}_q$ be obtained from $\mathbf{P}_0$, X_0,

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A_0, D_0 by extension of scalars. Diagram (5.1.1) of (5.6) comes from an analogous diagram over $\mathbf{F}_q$:

(6.1.1)
$$\begin{array}{ccc} X_0 & \xleftarrow{\pi_0} & \widetilde{X}_0 \\ & & \downarrow f_0 \\ & & D_0 \end{array}$$

Suppose that X is connected of *even* dimension $n+1 = 2m+2$, and that the pencil $(X_t)_{t\in D}$ of hyperplane sections of X defined by D is a *Lefschetz pencil*. The set S of $t \in D$ for which X_t is singular is defined over $\mathbf{F}_q$, i.e. comes from $S_0 \subset D_0$. Put $U_0 = D_0 - S_0$ and $U = D - S$.

Let $u \in U$. The vanishing part $E \subset H^n(X_u, \mathbf{Q}_\ell)$ of the cohomology is stable under $\pi_1(U, u)$, and therefore defines on U a local subsystem $\mathcal{E}$ of $R^n f_* \mathbf{Q}_\ell$. The latter is defined over $\mathbf{F}_q$: $R^i f_* \mathbf{Q}_\ell$ is the inverse image of the $\mathbf{Q}_\ell$-sheaf $R^i f_{0*} \mathbf{Q}_\ell$ on D_0, and, on U, $\mathcal{E}$ is the inverse image of a local subsystem

$$\mathcal{E}_0 \subset R^n f_{0*} \mathbf{Q}_\ell.$$

The cup product is an alternating form

$$\psi : R^n f_{0*} \mathbf{Q}_\ell \otimes R^n f_{0*} \mathbf{Q}_\ell \longrightarrow \mathbf{Q}_\ell(-n).$$

Let $\mathcal{E}_0^\perp$ denote the orthogonal complement of $\mathcal{E}_0$ with respect to ψ. On $R^n f_{0*} \mathbf{Q}_\ell | U_0$, one sees that ψ induces a perfect duality

$$\psi : \mathcal{E}_0/(\mathcal{E}_0 \cap \mathcal{E}_0^\perp) \otimes \mathcal{E}_0/(\mathcal{E}_0 \cap \mathcal{E}_0^\perp) \longrightarrow \mathbf{Q}_\ell(-n).$$

Theorem **(6.2)**. — *For every $x \in |U_0|$, the polynomial* $\det(1 - F_x^* t, \mathcal{E}_0/(\mathcal{E}_0 \cap \mathcal{E}_0^\perp))$ *has rational coefficients.*

Corollary **(6.3)**. — *Let j_0 be the inclusion of* U_0 *in* D_0*, and j that of* U *in* D*. The eigenvalues of* F^* *acting on* $H^1(D, j_* \mathcal{E}/(\mathcal{E} \cap \mathcal{E}^\perp))$ *are algebraic numbers all of whose complex conjugates α satisfy*

$$q^{\frac{n+1}{2} - \frac{1}{2}} \leq |\alpha| \leq q^{\frac{n+1}{2} + \frac{1}{2}}.$$

Indeed, by (5.10) and (6.2), the hypotheses of (3.2) are satisfied by $(U_0, \mathcal{E}_0/(\mathcal{E}_0 \cap \mathcal{E}_0^\perp), \psi)$, with $\beta = n$, and one applies (3.9).

Lemma **(6.4)**. — *Let $\mathcal{G}_0$ be a twisted constant $\mathbf{Q}_\ell$-sheaf on* U_0 *whose inverse image $\mathcal{G}$ on* U *is a constant sheaf. Then there exist ℓ-adic units α_i in $\overline{\mathbf{Q}}_\ell$ such that, for every $x \in |U_0|$,*

$$\det(1 - F_x^* t, \mathcal{G}_0) = \prod_i (1 - \alpha_i^{\deg(x)} t).$$

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This lemma says that $\mathcal{G}_0$ is the inverse image of a sheaf on $\mathrm{Spec}(\mathbf{F}_q)$, namely its direct image on $\mathrm{Spec}(\mathbf{F}_q)$. The latter identifies with an ℓ-adic representation G_0 of $\mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$, and we take

$$\det(1-\mathrm{F}t,\mathrm{G}_0)=\prod_i(1-\alpha_i t).$$

Lemma (6.4) applies to $\mathrm{R}^i f_{0*}\mathbf{Q}_\ell (i\neq n)$, to $\mathrm{R}^n f_{0*}\mathbf{Q}_\ell/\mathcal{E}_0$, and to $\mathcal{E}_0\cap\mathcal{E}_0^\perp$.

For $x\in|\mathrm{U}_0|$, the fiber $\mathrm{X}_x=f_0^{-1}(x)$ is a variety over the finite field $k(x)$. If $\overline{x}$ is a point of U above x, $\mathrm{X}_{\overline{x}}$ is obtained from X_x by extension of scalars from $k(x)$ to its algebraic closure $k(\overline{x})=\overline{\mathbf{F}}_q$, and $\mathrm{H}^i(\mathrm{X}_{\overline{x}},\mathbf{Q}_\ell)$ is the fiber of $\mathrm{R}^i f_*\mathbf{Q}_\ell$ at $\overline{x}$. Formula (1.5.4) for the variety X_x over $k(x)$ therefore reads

$$\mathrm{Z}(\mathrm{X}_x,t)=\prod_i\det(1-\mathrm{F}_x^*t,\mathrm{R}^i f_{0*}\mathbf{Q}_\ell)^{(-1)^{i+1}},$$

and $\mathrm{Z}(\mathrm{X}_x,t)$ is the product of

$$\begin{aligned}\mathrm{Z}^f&=\det(1-\mathrm{F}_x^*t,\mathrm{R}^n f_{0*}\mathbf{Q}_\ell/\mathcal{E}_0)\cdot\det(1-\mathrm{F}_x^*t,\mathcal{E}_0\cap\mathcal{E}_0^\perp)\\&\quad\cdot\prod_{i\neq n}\det(1-\mathrm{F}_x^*t,\mathrm{R}^i f_{0*}\mathbf{Q}_\ell)^{(-1)^{i+1}}\end{aligned}$$

and

$$\mathrm{Z}^m=\det(1-\mathrm{F}_x^*t,\mathcal{E}_0/(\mathcal{E}_0\cap\mathcal{E}_0^\perp)).$$

Put $\mathcal{F}_0=\mathcal{E}_0/(\mathcal{E}_0\cap\mathcal{E}_0^\perp),\mathcal{F}=\mathcal{E}/(\mathcal{E}\cap\mathcal{E}^\perp)$, and apply (6.4) to the factors of Z^f. We find that there exist ℓ-adic units $\alpha_i(1\leq i\leq\mathrm{N})$ and $\beta_j(1\leq j\leq\mathrm{M})$ in $\overline{\mathbf{Q}}_\ell$ such that for every $x\in|\mathrm{U}_0|$

$$\mathrm{Z}(\mathrm{X}_x,t)=\frac{\prod_i(1-\alpha_i^{\deg(x)}t)}{\prod_j(1-\beta_j^{\deg(x)}t)}\cdot\det(1-\mathrm{F}_x^*t,\mathcal{F}_0),$$

and in particular the right-hand side belongs to $\mathbf{Q}(t)$. If an α_i coincides with a β_j, it is permissible to delete simultaneously that α_i from the list of the $\alpha's$ and that β_j from the list of the $\beta's$. We may therefore suppose, and shall suppose, that $\alpha_i\neq\beta_j$ for every i and j.

(6.5) It is enough to prove that the polynomials $\prod_i(1-\alpha_i t)$ and $\prod_j(1-\beta_j t)$ have rational coefficients, i.e. that the family of the α_i (respectively, the family of the β_j) is defined over $\mathbf{Q}$. We shall deduce this from the following propositions.

Proposition **(6.6)**. — *Let $(\gamma_i)(1\leq i\leq\mathrm{P})$ and $(\delta_j)(1\leq j\leq\mathrm{Q})$ be two families of ℓ-adic units in $\overline{\mathbf{Q}}_\ell$. Suppose that $\gamma_i\neq\delta_j$. If* K *is a sufficiently large finite set of integers different from 1, and* L *is a sufficiently large subset of $|\mathrm{U}_0|$ of density 0, then, whenever $x\in|\mathrm{U}_0|$ satisfies $k\nmid\deg(x)$ (for every $k\in\mathrm{K}$) and $x\notin\mathrm{L}$, the denominator of*

(6.6.1) $$\det(1-\mathrm{F}_x^*t,\mathcal{F}_0)\cdot\prod_i(1-\gamma_i^{\deg(x)}t)/\prod_j(1-\delta_j^{\deg(x)}t),$$

when written in lowest terms, is $\prod_j(1-\delta_j^{\deg(x)}t)$.

The proof will be given in (6.10-13). By (6.7) below, (6.6) gives an intrinsic description of the family of the δ_j in terms of the family of rational functions (6.6.1), for $x \in |\mathrm{U}_0|$.

Lemma **(6.7).** — *Let* K *be a finite set of integers different from 1, and let* $(\delta_j)(1 \leq j \leq \mathrm{Q})$ *and* $(\varepsilon_j)(1 \leq j \leq \mathrm{Q})$ *be two families of elements of a field. If, for every sufficiently large n divisible by none of the* $k \in \mathrm{K}$, *the family of the* δ_j^n *coincides with that of the* ε_j^n *(up to order), then the family of the* δ_j *coincides with that of the* ε_j *(up to order).*

We argue by induction on Q. The set of integers n such that $\delta_{\mathrm{Q}}^n = \varepsilon_j^n$ is an ideal (n_j). Let us prove that there is a j_0 such that $\delta_{\mathrm{Q}} = \varepsilon_{j_0}$. Otherwise the n_j would all be different from 1, and there would be arbitrarily large integers n divisible by none of the n_j and by none of the $k \in \mathrm{K}$. We would then have $\delta_{\mathrm{Q}}^n \neq \varepsilon_j^n$, contradicting the hypothesis. Thus there is a j_0 such that $\delta_{\mathrm{Q}} = \varepsilon_{j_0}$. The conclusion follows by applying the induction hypothesis to the families $(\delta_j)(j \neq \mathrm{Q})$ and $(\varepsilon_j)(j \neq j_0)$.

Proposition **(6.8).** — *Let* $(\gamma_i)(1 \leq i \leq \mathrm{P})$ *and* $(\delta_j)(1 \leq j \leq \mathrm{Q})$ *be two families of p-adic units in* $\overline{\mathbf{Q}}_\ell$, *and put* $\mathrm{R}(t) = \prod_i (1 - \gamma_i t)$ *and* $\mathrm{S}(t) = \prod_j (1 - \delta_j t)$. *Suppose that for every* $x \in |\mathrm{U}_0|$,

$$\prod_i (1 - \delta_i^{\deg(x)} t) \text{ divides}$$

$$\prod_i (1 - \gamma_i^{\deg(x)} t) \cdot \det(1 - \mathrm{F}_x^* t, \mathcal{F}_0).$$

Then $\mathrm{S}(t)$ *divides* $\mathrm{R}(t)$.

Remove from the families (γ_i) and (δ_j) pairs of common elements until the hypothesis of (6.6) is satisfied. Apply (6.6). By hypothesis, the rational functions (6.6.1) are polynomials. No δ therefore remains, which means that $\mathrm{S}(t)$ divides $\mathrm{R}(t)$.

This proposition gives an intrinsic characterization of $\mathrm{R}(t)$ in terms of the family of polynomials

$$\prod_i (1 - \gamma_i^{\deg(x)} t) \cdot \det(1 - \mathrm{F}_x^* t, \mathcal{F}_0).$$

It is the least common multiple of the polynomials $\mathrm{S}(t) = \prod_j (1 - \delta_j t)$ satisfying the hypothesis of (6.8).

(6.9) Let us prove (6.5), and hence (6.2), modulo (6.6). In (6.6), take $(\gamma_i) = (\alpha_i)$ and $(\delta_j) = (\beta_j)$. We obtain an intrinsic characterization of the family of the β_j in terms of the family of rational functions $\mathrm{Z}(\mathrm{X}_x, t)(x \in |\mathrm{U}_0|)$. Since these belong to $\mathbf{Q}(t)$, the family of the β_j is defined over $\mathbf{Q}$.

The polynomials $\prod_i (1 - \alpha_i^{\deg(x)} t) \cdot \det(1 - \mathrm{F}_x^* t, \mathcal{F}_0)$ therefore belong to $\mathbf{Q}[t]$. Proposition (6.8) gives an intrinsic description of the family of the α_i in terms of this family of polynomials. The family of the α_i is therefore defined over $\mathbf{Q}$.

(6.10) *Preliminaries.* — Let $u \in \mathrm{U}$, and let $\mathcal{F}_u$ be the fiber of $\mathcal{F}$ at u. The arithmetic fundamental group $\pi_1(\mathrm{U}_0, u)$, an extension of $\hat{\mathbf{Z}} = \mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ (generator: φ) by the geometric fundamental group $\pi_1(\mathrm{U}, u)$, acts on $\mathcal{F}_u$ by symplectic similitudes:

$$\rho : \pi_1(\mathrm{U}_0, u) \longrightarrow \mathrm{CSp}(\mathcal{F}_u, \psi).$$

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We denote by $\mu(g)$ the multiplier of a symplectic similitude g. Let

$$\mathrm{H} \subset \hat{\mathbf{Z}} \times \mathrm{CSp}(\mathcal{F}_u, \psi)$$

be the subgroup defined by the equation

$$q^{-n} = \mu(g)$$

(q being an ℓ-adic unit, $q^n \in \mathbf{Q}_\ell^*$ is defined for every $n \in \hat{\mathbf{Z}}$). The fact that ψ takes values in $\mathbf{Q}_\ell(-n)$ is expressed by saying that the map from π_1 to $\hat{\mathbf{Z}} \times \mathrm{CSp}$, whose components are the canonical projection onto $\hat{\mathbf{Z}}$ and ρ, factors through

$$\rho_1 : \pi_1(\mathrm{U}_0, u) \longrightarrow \mathrm{H}.$$

Lemma **(6.11)**. — *The image* H_1 *of* ρ_1 *is open in* H.

Indeed, $\pi_1(\mathrm{U}_0, u)$ maps onto $\hat{\mathbf{Z}}$, and the image of $\pi_1(\mathrm{U}, u) = \mathrm{Ker}(\pi_1(\mathrm{U}_0, u) \longrightarrow \hat{\mathbf{Z}})$ in $\mathrm{Sp}(\mathcal{F}_u, \psi) = \mathrm{Ker}(\mathrm{H} \longrightarrow \hat{\mathbf{Z}})$ is open by (5.10).

Lemma **(6.12)**. — *For an ℓ-adic unit $\delta \in \overline{\mathbf{Q}}_\ell$, the set* Z *of $(n, g) \in \mathrm{H}_1$ such that δ^n is an eigenvalue of g is a closed set of measure zero.*

It is clear that Z is closed. For every $n \in \hat{\mathbf{Z}}$, let CSp_n be the set of $g \in \mathrm{CSp}(\mathcal{F}_u, \psi)$ such that $\mu(g) = q^{-n}$, and let Z_n be the set of $g \in \mathrm{CSp}_n$ such that δ^n is an eigenvalue of g. Then CSp_n is a homogeneous space under Sp, and one checks that Z_n is a proper algebraic subspace, hence has measure zero. By (6.11), $\mathrm{H}_1 \cap (\{n\} \times \mathrm{Z}_n)$ therefore has measure zero in the inverse image of n in H_1, and Fubini is applied to the projection $\mathrm{H}_1 \longrightarrow \hat{\mathbf{Z}}$.

(6.13) Let us prove (6.6). For every i and j, the set of integers n such that $\gamma_i^n = \delta_j^n$ is the set of multiples of a fixed integer n_{ij} (we do not exclude $n_{ij} = 0$). By hypothesis, $n_{ij} \neq 1$.

By (6.12) and the Chebotarev density theorem, the set of $x \in |\mathrm{U}_0|$ such that some $\beta_j^{\deg(x)}$ is an eigenvalue of F_x^* acting on $\mathcal{F}_0$ has density zero. Take for K the set of the n_{ij}, and for L the set of x just described.

7. End **of the proof of 1.7.**

Lemma **(7.1)**. — *Let* X_0 *be a smooth absolutely irreducible projective variety of* even *dimension d over* $\mathbf{F}_q$. *Let* X *over* $\overline{\mathbf{F}}_q$ *be obtained from* X_0 *by extension of scalars, and let α be an eigenvalue of* F^* *acting on* $\mathrm{H}^d(\mathrm{X}, \mathbf{Q}_\ell)$. *Then α is an algebraic number all of whose complex conjugates, again denoted α, satisfy*

(7.1.1) $$q^{\frac{d}{2}-\frac{1}{2}} \leq |\alpha| \leq q^{\frac{d}{2}+\frac{1}{2}}.$$

We argue by induction on d (always assumed even). The case $d = 0$ is trivial, even without assuming X_0 absolutely irreducible; henceforth suppose $d \geq 2$. Put $d = n + 1 = 2m + 2$.

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If $\mathbf{F}_{q'}$ is an extension of degree r of $\mathbf{F}_q$ and $X'_0/\mathbf{F}_{q'}$ is obtained from $X_0/\mathbf{F}_q$ by extension of scalars, assertion (7.1) for $X_0/\mathbf{F}_q$ is equivalent to (7.1) for $X'_0/\mathbf{F}_{q'}$: just as q is replaced by q^r, the eigenvalues of F^* are replaced by their r-th powers.

By (5.7), in a suitable projective embedding $i : X \longrightarrow \mathbf{P}$, X admits a Lefschetz pencil of hyperplane sections. The preceding remark allows us to suppose that i and this pencil are defined over $\mathbf{F}_q$ (after replacing $\mathbf{F}_q$ by a finite extension if necessary).

Suppose, then, that over $\mathbf{F}_q$ there is a projective embedding $X_0 \longrightarrow \mathbf{P}_0$ and a codimension-two linear subspace A_0 of $\mathbf{P}_0$ defining a Lefschetz pencil. We retain the notation of (6.1) and (6.3). A further extension of scalars reduces us to assuming that:

a) The points of S are defined over $\mathbf{F}_q$.

b) The vanishing cycles at $x_s (s \in S)$ are defined over $\mathbf{F}_q$ (since only $\pm\delta$ is intrinsic, they might otherwise be defined only over a quadratic extension).

c) There is a rational point $u_0 \in U_0$. We take the corresponding point u of U as base point.

d)$X_{u_0} = f_0^{-1}(u_0)$ admits a smooth hyperplane section Y_0 defined over $\mathbf{F}_q$. Put $Y = Y_0 \otimes_{\mathbf{F}_q} \overline{\mathbf{F}}_q$.

Since $\widetilde{X}$ is obtained from X by blowing up the smooth codimension-two subvariety $A \cap X$, one has

$$H^i(X, \mathbf{Q}_\ell) \hookrightarrow H^i(\widetilde{X}, \mathbf{Q}_\ell)$$

(in fact, $H^i(\widetilde{X}, \mathbf{Q}_\ell) = H^i(X, \mathbf{Q}_\ell) \oplus H^{i-2}(A \cap X, \mathbf{Q}_\ell)(-1)$). It is therefore enough to prove (7.1.1) for the eigenvalues α of F^* acting on $H^d(\widetilde{X}, \mathbf{Q}_\ell)$.

The Leray spectral sequence for f is

$$E_2^{pq} = H^p(D, R^q f_* \mathbf{Q}_\ell) \Rightarrow H^{p+q}(\widetilde{X}, \mathbf{Q}_\ell).$$

It is enough to prove (7.1.1) for the eigenvalues α of F^* acting on E_2^{pq} with $p + q = d = n + 1$. These are:

A)$E_2^{2,n-1}$. By (5.8), $R^{n-1} f_* \mathbf{Q}_\ell$ is constant. By (2.10), therefore,

$$E_2^{2,n-1} = H^{n-1}(X_u, \mathbf{Q}_\ell)(-1).$$

By the weak Lefschetz theorem (a corollary of SGA 4, XIV (3.2), and Poincaré duality, SGA 4, XVIII), one has

$$H^{n-1}(X_u, \mathbf{Q}_\ell)(-1) \hookrightarrow H^{n-1}(Y, \mathbf{Q}_\ell)(-1),$$

and the induction hypothesis is applied to Y_0.

B)$E_2^{0,n+1}$. If the vanishing cycles are nonzero, $R^{n+1} f_* \mathbf{Q}_\ell$ is constant and

$$E_2^{0,n+1} = H^{n+1}(X_u, \mathbf{Q}_\ell).$$

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The Gysin morphism

$$\mathrm{H}^{n-1}(\mathrm{Y}, \mathbf{Q}_\ell)(-1) \longrightarrow \mathrm{H}^{n+1}(\mathrm{X}_u, \mathbf{Q}_\ell)$$

is surjective (this argument is dual to A)), and the induction hypothesis is applied to Y_0.
If the vanishing cycles are zero, the exact sequence of $(5.8)b)$ gives an exact sequence

$$\bigoplus_{s\in \mathrm{S}} \mathbf{Q}_\ell(m-n) \longrightarrow \mathrm{E}_2^{0,n+1} \longrightarrow \mathrm{H}^{n+1}(\mathrm{X}_u, \mathbf{Q}_\ell).$$

The eigenvalue of F acting on $\mathbf{Q}_\ell(m-n)$ is $q^{d/2}$, and H^{n+1} is handled as above.
C)$\mathrm{E}_2^{1,n}$. If the "hard" Lefschetz theorem were available, we would know that $\mathcal{E} \cap \mathcal{E}^\perp$ is zero and that $\mathrm{R}^n f_* \mathbf{Q}_\ell$ is the direct sum of $j_*\mathcal{E}$ and a constant sheaf. The H^1 of a constant sheaf on $\mathbf{P}^1$ is zero, and it would be enough to apply (6.3).
Since we have not yet proved "hard" Lefschetz, we shall have to proceed by dévissage. If the vanishing cycles are zero, $\mathrm{R}^n f_* \mathbf{Q}_\ell$ is constant by $(5.8)b)$, and $\mathrm{E}_2^{1,n} = 0$. We may therefore suppose, and shall suppose, that the vanishing cycles are nonzero. Filter $\mathrm{R}^n f_* \mathbf{Q}_\ell = j_* j^* \mathrm{R}^n f_* \mathbf{Q}_\ell (5.8)$ by the subsheaves $j_*\mathcal{E}$ and $j_*(\mathcal{E} \cap \mathcal{E}^\perp)$. If the vanishing cycles δ do not lie in $\mathcal{E} \cap \mathcal{E}^\perp$, there are exact sequences:

(7.1.2) $$0 \longrightarrow j_*\mathcal{E} \longrightarrow \mathrm{R}^n f_* \mathbf{Q}_\ell \longrightarrow \text{ constant sheaf } \longrightarrow 0.$$

(7.1.3) $$\begin{gathered} 0 \longrightarrow \text{ the constant sheaf } j_*(\mathcal{E} \cap \mathcal{E}^\perp) \longrightarrow j_*\mathcal{E} \\ \longrightarrow j_*(\mathcal{E}/(\mathcal{E} \cap \mathcal{E}^\perp)) \longrightarrow 0. \end{gathered}$$

If, God forbid, the δ lie in $\mathcal{E} \cap \mathcal{E}^\perp$, then $\mathcal{E} \subset \mathcal{E}^\perp$, and there are exact sequences:

(7.1.4) $$0 \longrightarrow \text{ the constant sheaf } j_*\mathcal{E}^\perp \longrightarrow \mathrm{R}^n f_* \mathbf{Q}_\ell \longrightarrow \text{ a sheaf } \mathcal{F} \longrightarrow 0,$$

(7.1.5) $$\begin{gathered} 0 \longrightarrow \mathcal{F} \longrightarrow \text{ the constant sheaf } j_* j^* \mathcal{F} \\ \longrightarrow \bigoplus_{s\in \mathrm{S}} \mathbf{Q}_\ell(n-m)_s \longrightarrow 0. \end{gathered}$$

In the first case, the long exact cohomology sequences give

(7.1.2′) $$\mathrm{H}^1(\mathrm{D}, j_*\mathcal{E}) \longrightarrow \mathrm{H}^1(\mathrm{R}^n f_* \mathbf{Q}_\ell) \longrightarrow 0,$$

(7.1.3′) $$0 \longrightarrow \mathrm{H}^1(\mathrm{D}, j_*\mathcal{E}) \longrightarrow \mathrm{H}^1(\mathrm{D}, j_*(\mathcal{E}/(\mathcal{E} \cap \mathcal{E}^\perp))),$$

and one applies (6.3).
In the second case, they give

(7.1.4′) $$0 \longrightarrow \mathrm{H}^1(\mathrm{D}, \mathrm{R}^n f_* \mathbf{Q}_\ell) \longrightarrow \mathrm{H}^1(\mathrm{D}, \mathcal{F}),$$

(7.1.5′) $$\bigoplus_{s\in \mathrm{S}} \mathbf{Q}_\ell(n-m) \longrightarrow \mathrm{H}^1(\mathrm{D}, \mathcal{F}) \longrightarrow 0,$$

and one notes that F acts on $\mathbf{Q}_\ell(n-m)$ by multiplication by $q^{d/2}$.
Lemma **(7.2).** — *Let* X_0 *be a smooth absolutely irreducible projective variety of dimension* d *over* $\mathbf{F}_q$. *Let* X *over* $\overline{\mathbf{F}}_q$ *be obtained from* X_0 *by extension of scalars, and let* α *be an eigenvalue of* F^* *acting on* $\mathrm{H}^d(\mathrm{X}, \mathbf{Q}_\ell)$. *Then* α *is an algebraic number all of whose complex conjugates, again denoted* α, *satisfy*

$$|\alpha| = q^{d/2}.$$

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Let us first prove that (7.2) implies (1.7). For X_0 smooth and projective over $\mathbf{F}_q$ and i an integer, the following assertion must be proved:

$\mathrm{W}(\mathrm{X}_0, i)$. *Let* X *be obtained from* X_0 *by extension of scalars from* $\mathbf{F}_q$ *to* $\overline{\mathbf{F}}_q$. *If* α *is an eigenvalue of* F^* *acting on* $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$*, then* α *is an algebraic number all of whose complex conjugates, again denoted* α*, satisfy* $|\alpha| = q^{i/2}$.

a) If $\mathbf{F}_{q^n}$ is an extension of degree n of $\mathbf{F}_q$, and $\mathrm{X}_0'/\mathbf{F}_{q^n}$ is obtained from $\mathrm{X}_0/\mathbf{F}_q$ by extension of scalars, then $\mathrm{W}(\mathrm{X}_0, i)$ is equivalent to $\mathrm{W}(\mathrm{X}_0', i)$: extending the scalars amounts to replacing α by α^n and q by q^n.

b) If X_0 is pure of dimension n, $\mathrm{W}(\mathrm{X}_0, i)$ is equivalent to $\mathrm{W}(\mathrm{X}_0, 2n-i)$: this follows from Poincaré duality.

c) If X_0 is a sum of varieties X_0^α, $\mathrm{W}(\mathrm{X}_0, i)$ is equivalent to the conjunction of the $\mathrm{W}(\mathrm{X}_0^\alpha, i)$.

d) If X_0 is pure of dimension n, Y_0 is a smooth hyperplane section of X_0, and $i < n$, then $\mathrm{W}(\mathrm{Y}_0, i)$ implies $\mathrm{W}(\mathrm{X}_0, i)$: this follows from the weak Lefschetz theorem.

To prove the assertions $\mathrm{W}(\mathrm{X}_0, i)$, we reduce successively:

— by *c*), to the case where X_0 is pure of some dimension n;

— by *b*), to the further case $0 \le i \le n$;

— by a) and *d*), to the further case $i = n$;

— by a) and *c*), to the further case where X_0 is absolutely irreducible.

This case is covered by (7.2).

(7.3) Let us prove (7.2). For every integer k, α^k is an eigenvalue of F^* acting on $\mathrm{H}^{kd}(\mathrm{X}^k, \mathbf{Q}_\ell)$ (Künneth formula). For k even, X^k is covered by (7.1), whence

$$q^{\frac{kd}{2}-\frac{1}{2}} \le |\alpha^k| \le q^{\frac{kd}{2}-\frac{1}{2}}$$

and

$$q^{\frac{d}{2}-\frac{1}{2k}} \le |\alpha| \le q^{\frac{d}{2}+\frac{1}{2k}}.$$

Letting k tend to infinity gives (7.2).

8. First applications.

Theorem **(8.1)**. — *Let* $\mathrm{X}_0 \subset \mathbf{P}_0^{n+r}$ *be a smooth complete intersection over* $\mathbf{F}_q$*, of dimension* n *and multidegree* $(d_1, \cdots, d_r)$*. Let* b' *be the* n*-th Betti number of smooth complex complete intersections of the same dimension and multidegree. Put* $b = b'$ *for* n *odd and* $b = b' - 1$ *for* n *even. Then*

$$|\#\mathrm{X}_0(\mathbf{F}_q) - \#\mathbf{P}^n(\mathbf{F}_q)| \le b \cdot q^{n/2}.$$

Let $\mathrm{X}/\overline{\mathbf{F}}_q$ be obtained from X_0, and let $\mathbf{Q}_\ell \cdot \eta^i$ be the line in $\mathrm{H}^{2i}(\mathrm{X}, \mathbf{Q}_\ell)$ generated by the i-th cup power of the cohomology class of a hyperplane section. On this line F^* acts by multiplication by q^i. The cohomology of X is the sum of the $\mathbf{Q}_\ell \cdot \eta^i$

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$(0 \le i \le n)$ and of the primitive part of $\mathrm{H}^n(\mathrm{X}, \mathbf{Q}_\ell)$, of dimension b. By (1.5), there therefore exist b algebraic numbers α_j, the eigenvalues of F^* acting on this primitive cohomology, such that

$$\#\mathrm{X}_0(\mathbf{F}_q) = \sum_{i=0}^{n} q^i + (-1)^n \sum_j \alpha_j.$$

By (1.7), $|\alpha_j| = q^{n/2}$ and

$$|\#\mathrm{X}_0(\mathbf{F}_q) - \#\mathbf{P}(\mathbf{F}_q)| = |\#\mathrm{X}_0(\mathbf{F}_q) - \sum_{i=0}^{n} q^i| = |\sum_j \alpha_j| \le \sum_j |\alpha_j| = b \cdot q^{n/2}.$$

Theorem **(8.2)**. — *Let* N *be an integer* ≥ 1, *let* $\varepsilon : (\mathbf{Z}/\mathrm{N})^* \longrightarrow \mathbf{C}^*$ *be a character, let* k *be an integer* ≥ 2, *and let* f *be a holomorphic modular form on* $\Gamma_0(\mathrm{N})$, *of weight* k *and character* ε : f *is a holomorphic function on the Poincaré upper half-plane* X *such that, for* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbf{Z})$, *with* $c \equiv 0(\mathrm{N})$, *one has*

$$f\left(\frac{az+b}{cz+d}\right) = \varepsilon(a)^{-1}(cz+d)^k f(z).$$

Assume that f *is* cuspidal *and* primitive *("new" in the sense of Atkin-Lehner and Miyake), and in particular an eigenvector of the Hecke operators* $\mathrm{T}_p (p \nmid \mathrm{N})$. *Set* $f = \sum_{n=1}^{\infty} a_n q^n$, *with* $q = e^{2\pi i z}$ *(and* $a_1 = 1$*). Then, for every prime* p *not dividing* N,

$$|a_p| \le 2 \cdot p^{\frac{k-1}{2}}.$$

In other words, the roots of the equation

$$\mathrm{T}^2 - a_p \mathrm{T} + \varepsilon(p) \cdot p^{k-1}$$

have absolute value $p^{\frac{k-1}{2}}$.

Indeed, these roots are eigenvalues of Frobenius acting on H^{k-1} of a nonsingular projective variety of dimension $k-1$ defined over $\mathbf{F}_p$.

Under restrictive hypotheses, this fact is proved in my Bourbaki seminar talk (Formes modulaires et représentations ℓ-adiques, exposé 355, February 1969, in: *Lecture Notes in Mathematics*, 179). The general case is not much more difficult.

Remark **(8.3)**. — J. P. Serre and I have recently proved that (8.2) remains true for $k = 1$. The proof is entirely different.

The following application was suggested to me by E. Bombieri.

Theorem **(8.4)**. — *Let* Q *be a polynomial in* n *variables, of degree* d, *over* $\mathbf{F}_q$; *let* Q_d *be the homogeneous part of degree* d *of* Q; *and let* $\psi : \mathbf{F}_q \longrightarrow \mathbf{C}^*$ *be a nontrivial additive character of* $\mathbf{F}_q$. *Assume that:*

(i) *d is relatively prime to p;*
(ii) *the hypersurface* H_0 *in* $\mathbf{P}^{n-1}_{\mathbf{F}_q}$ *defined by* Q_d *is smooth.*

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Then

$$|\sum_{x_1,\cdots,x_n\in\mathbf{F}_q}\psi(\mathrm{Q}(x_1,\cdots,x_n))|\leq(d-1)^n q^{n/2}.$$

Replacing Q by a scalar multiple if necessary, we may assume (and shall assume) that

(8.4.1)
$$\psi(x)=\exp(2\pi i\,\mathrm{Tr}_{\mathbf{F}_q/\mathbf{F}_p}(x)/p).$$

Let X_0 be the étale covering of the n-dimensional affine space A_0 over $\mathbf{F}_q$ with equation $\mathrm{T}^p-\mathrm{T}=\mathrm{Q}$, and let σ be the projection from X_0 to A_0:

$$\sigma:\mathrm{X}_0\longrightarrow\mathrm{A}_0$$

$$\mathrm{X}_0=\mathrm{Spec}(\mathbf{F}_q[x_1,\cdots,x_n,\mathrm{T}]/(\mathrm{T}^p-\mathrm{T}-\mathrm{Q})).$$

The covering X_0 is Galois, with Galois group $\mathbf{Z}/p$; $i\in\mathbf{Z}/p=\mathbf{F}_p$ acts by $\mathrm{T}\longmapsto\mathrm{T}+i$.

Let $x\in\mathrm{A}_0(\mathbf{F}_q)$, and let us compute the Frobenius endomorphism of the fiber of $\mathrm{X}_0/\mathrm{A}_0$ over x. Set $q=p^f$, and let $\overline{\mathbf{F}}_q$ be an algebraic closure of $\mathbf{F}_q$. For $(x,\mathrm{T})\in\mathrm{X}_0(\overline{\mathbf{F}}_q)$ lying over x, we have $\mathrm{F}((x,\mathrm{T}))=(x,\mathrm{T}^q)$, and

$$\mathrm{T}^q=\mathrm{T}+\sum_{i=1}^{f}(\mathrm{T}^{p^i}-\mathrm{T}^{p^{i-1}})=\mathrm{T}+\sum_i\mathrm{Q}(x)^{p^{i-1}}=\mathrm{T}+\mathrm{Tr}_{\mathbf{F}_q/\mathbf{F}_p}(\mathrm{Q}(x)).$$

Thus Frobenius acts as the element $\mathrm{Tr}_{\mathbf{F}_q/\mathbf{F}_p}(\mathrm{Q}(x))$ of the Galois group.

Let E be the field of the p-th roots of unity and let λ be a finite place of E not dividing p. We shall work in λ-adic cohomology. For $j\in\mathbf{Z}/p$, let $\mathcal{F}_{j,0}$ be the rank-one E_λ-local system on A_0 defined by X_0 and $\psi(-jx):\mathbf{Z}/p\longrightarrow\mathrm{E}^*\longrightarrow\mathrm{E}_\lambda^*$: there is a map $\iota:\mathrm{X}_0\longrightarrow\mathcal{F}_{j,0}$, and $\iota(i*x)=\psi(-ji)\iota(x)$. Denote without the subscript 0 the objects obtained from $\mathrm{A}_0,\mathrm{X}_0,\mathcal{F}_{j,0}$ by extension of scalars to $\overline{\mathbf{F}}_q$. The trace formula (1.12.1) for $\mathcal{F}_{j,0}$ reads

(8.4.2)
$$\sum_{x_1,\cdots,x_n\in\mathbf{F}_q}\psi(\mathrm{Q}(x_1,\cdots,x_n))=\sum_i(-1)^i\,\mathrm{Tr}(\mathrm{F}^*,\mathrm{H}^i_c(\mathrm{A},\mathcal{F}_1)).$$

We have $\sigma_*\mathrm{E}_\lambda=\bigoplus_j\mathcal{F}_j$, and hence

(8.4.3)
$$\mathrm{H}^*_c(\mathrm{X},\mathbf{Q}_\ell)\otimes_{\mathbf{Q}_\ell}\mathrm{E}_\lambda=\bigoplus_j\mathrm{H}^*_c(\mathrm{A},\mathcal{F}_j).$$

For $j=0$, $\mathcal{F}_j$ is the constant sheaf E_λ; this summand corresponds, via pullback, to the inclusion of the cohomology of A into that of X.

Lemma **(8.5)**. — (i) *For $j\neq0$, $\mathrm{H}^i_c(\mathrm{A},\mathcal{F}_j)$ vanishes for $i\neq n$; for $i=n$, this cohomology space has dimension $(d-1)^n$.*

(ii) *For $j\neq0$, the cup product*

$$\mathrm{H}^n_c(\mathrm{A},\mathcal{F}_j)\otimes\mathrm{H}^n_c(\mathrm{A},\mathcal{F}_{-j})\longrightarrow\mathrm{H}^{2n}_c(\mathrm{A},\mathrm{E}_\lambda)\xrightarrow{\mathrm{Tr}}\mathrm{E}_\lambda(-n)$$

is a perfect pairing.

(iii) X_0 *is an open subvariety of a nonsingular projective variety* Z_0.

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Let us deduce (8.4) from (8.5). Let $j_0 : X_0 \hookrightarrow Z_0$, and let $j : X \hookrightarrow Z$ be the map obtained from it by extension of scalars to $\overline{\mathbf{F}}_q$. By (8.4.2), (i), and (1.7) applied to Z_0, it suffices to prove the injectivity of

$$H^n_c(A, \mathcal{F}_1) \xrightarrow{\sigma^*} H^n_c(X, \mathcal{F}_1) = H^n_c(X, E_\lambda) \xrightarrow{j_!} H^n(Z, E_\lambda).$$

We have $\mathrm{Tr}(a \cup b) = \frac{1}{p}\,\mathrm{Tr}(j_!\sigma^* a \cup j_!\sigma^* b)$, and hence this injectivity follows from (ii).

(8.6) Let us prove (8.5) (iii). Let P_0 be the projective space over $\mathbf{F}_q$ obtained from A_0 by adjoining a hyperplane at infinity P_0^∞; let $H_0 \subset P_0^\infty$ have equation $Q_d = 0$; and let Y_0 be the covering of P_0 obtained by normalizing P_0 in X_0.

(8.6.1)
$$\begin{array}{ccccccc} X_0 & \hookrightarrow & Y_0 & & & & \\ \sigma\downarrow & & \downarrow & & & & \\ A_0 & \hookrightarrow & P_0 & \hookleftarrow & P_0^\infty & \hookleftarrow & H_0 \end{array}$$

Let us study Y_0/P_0 near infinity, étale-locally.

Lemma **(8.7)**. — Y_0 *is smooth away from the inverse image of* H_0.

The divisor of the rational function Q on P_0 is the sum of its finite part $\mathrm{div}(Q)_f$ and ($-d$) times the hyperplane at infinity. We have:

(8.7.1)
$$\mathrm{div}(Q) = \mathrm{div}(Q)_f - dP_0^\infty$$
$$\mathrm{div}(Q)_f \cap P_0^\infty = H_0.$$

On the affine part, $Y_0 = X_0$ is étale over A_0 and hence smooth. At infinity, but away from the inverse image of H_0, there are local coordinates $(z_1, \cdots, z_n)$ in which $Q = z_1^{-d}$ (this uses $(d,p) = 1$). In these coordinates, Y_0 is seen as the product of a curve and a smooth space (corresponding to the coordinates $z_2, \cdots, z_n$). By normality, it is smooth.

Lemma **(8.8)**. — *In an étale neighborhood of a point above* H_0, Y_0 *is smooth over one and the same normal singular surface.*

This time one can find local coordinates in which $Q = z_1^{-d} z_2$. Indeed, since H_0 is smooth, $\mathrm{div}(Q)_f$ is smooth near infinity and meets P_0^∞ transversely. This form is independent of the chosen point and involves only two coordinates, which proves the assertion.

(8.9) It is known that the following procedure (due to Zariski) resolves surface singularities: one alternately normalizes and blows up the (reduced) singular locus. The operations involved commute with étale localization and with taking a product with a smooth space. Zariski's procedure therefore also resolves the singularities of a space which, like Y_0, is étale-locally smooth over a surface. The resulting resolution of Y_0 is the desired Z_0.

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If T is a curve on a surface S containing the singular locus, and T′ is the inverse image of T in Zariski's resolution S′ of S, it is known that by iteratively blowing up in S′ the (reduced) singular locus of $(T')_{\mathrm{red}}$, one obtains a surface S″ such that the reduced inverse image $(T'')_{\mathrm{red}}$ of T in S″ is a normal crossings divisor. Once again, the operations involved commute with étale localization and with taking a product with a smooth space. Arguing as above, and observing that (Y_0, the locus at infinity) is locally smooth over a pair (S, T), one can even find Z_0 such that $Z_0 - X_0$ is a normal crossings divisor.

(8.10) Let us prove (8.5)(i), (ii). These are geometric assertions; this allows us henceforth to work over $\overline{\mathbf{F}}_q$. Let S′ be the affine space over $\overline{\mathbf{F}}_q$ that parametrizes polynomials in n variables of degree $\leq d$, and let S be the open subset of S′ corresponding to polynomials whose homogeneous part of degree d has nonzero discriminant. Let $Q_S \in H^0(S, \mathcal{O}_S[x_1, \cdots, x_n])$ denote the universal polynomial over S, and let X_S be the étale Galois covering of $A_S = A^n \times S$ with equation $T^p - T = Q_S$, with Galois group $\mathbf{Z}/p$. Let $P_S = \mathbf{P}^n \times S$ be the projective space over S completing A_S and let Y_S be the normalization of P_S in X_S. Over S, we have a diagram analogous to (8.6.1).

The expressions for Q in local coordinates given in (8.7) and (8.8) remain valid in the present situation, with parameters, so that, étale-locally on Y_S, Y_S/S is isomorphic to the product of S (which is smooth) with a fiber. The canonical resolution method used in (8.9) provides us with a relative compactification Z_S/S of X_S/S, with $Z_S - X_S a$ relative normal crossings divisor over S

$$\begin{array}{ccc} X_S & \xrightarrow{u} & Z_S \\ \sigma \downarrow & & \downarrow f \\ A_S & \xrightarrow{a} & S \end{array}$$

(f proper and smooth, u an open immersion, $Z_S - X_S a$ relative normal crossings divisor).

Let $\mathcal{F}_{j,S}$ be the E_λ-sheaf on A_S obtained as in (8.4) from X_S/A_S. We have $\sigma_* E_\lambda = \bigoplus_j \mathcal{F}_{j,S}$, hence

$$R^*(fu)_!(E_\lambda) = \bigoplus_j R^* a_! \mathcal{F}_{j,S}.$$

The properties of Z_S ensure that the $R^i(fu)_! E_\lambda = R^i f_*(u_! E_\lambda)$ are locally constant sheaves on S. Consequently, the $R^i a_! \mathcal{F}_{j,S}$ are also locally constant. Since S is connected, it suffices to prove (8.5)(i), (ii) for one particular polynomial Q. We shall take $Q = \sum_i x_i^d$. This polynomial satisfies the nonsingularity condition because $(d, p) = 1$. For this polynomial, the variables separate in the exponential sum of (8.4). This corresponds to the fact that $\mathcal{F}_j$ is the tensor product of the pullbacks of

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analogous sheaves $\mathcal{F}_j^1$ on the one-dimensional factors A^1 of $\mathrm{A} = \mathrm{A}^n$. By the Künneth formula

$$\mathrm{H}^*(\mathrm{A}, \mathcal{F}_j) = \bigotimes \mathrm{H}^*(\mathrm{A}^1, \mathcal{F}_j^1).$$

This reduces the proof of (8.5)(i), (ii) to the case where $n = 1$ and Q is x^d.

(8.11) Let us treat this special case. The covering X of A is irreducible, so that for $i = 0, 2$

$$\mathrm{H}_c^i(\mathrm{A}, \mathrm{E}_\lambda) \xrightarrow{\sim} \mathrm{H}_c^i(\mathrm{X}, \mathrm{E}_\lambda).$$

For $i \neq 1$ and $j \neq 0$, we therefore have

$$\mathrm{H}_c^i(\mathrm{A}, \mathcal{F}_j) = 0.$$

Assertion (ii) follows from (2.8) or (2.12) and from the fact that $u_!\mathcal{F}_j = u_*\mathcal{F}_j$. To prove ($i$), it remains to verify that

$$\chi_c(\mathrm{A}, \mathcal{F}_j) = 1 - d.$$

By the Euler-Poincaré formula (see Bourbaki seminar talk 286 of February 1965, by M. Raynaud), this is equivalent to the following lemma.

Lemma **(8.12)**. — *The Swan conductor of $\mathcal{F}_j$ at infinity is equal to d.*

This statement is equivalent to the following.

Lemma **(8.13)**. — *Let k be a finite field of characteristic p, let $y \in k[[x]]$ be an element of valuation d, with d relatively prime to p, let* L *be the extension of* $\mathrm{K} = k((x))$ *generated by the roots of* $\mathrm{T}^p - \mathrm{T} = y^{-1}$*, and let χ be the following character of* Gal(L/K) *with values in* $\mathbf{Z}/p$*:*

$$\chi(\sigma) = \sigma\mathrm{T} - \mathrm{T}.$$

Then, χ has conductor $d + 1$.

After extending the residue field, we reduce to the case where k is algebraically closed rather than finite, and apply: J. P. Serre, Sur les corps locaux à corps résiduel algébriquement clos, *Bull. Soc. Math. France*, 89 (1961), p. 105-154, no. 4.4.

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